

BUSINESS CONTINUITY PLAN (BCP)

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**BUSINESS CONTINUITY PLAN
(BCP)**

Q-Exel GXP, Pune

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APPROVAL

Business Continuity Plan (BCP) has been prepared, reviewed, approved and authorized by the following personnel.

	Name	Designation / Department	Signature / Date
Prepared by			
Reviewed by			
Reviewed by			
Approved by			
Authorized by			

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Executive Summary – Business Continuity Plan (BCP)

XXX (Firm Name) Limited stands at the forefront of Pharmaceutical and Biopharmaceutical landscape, with a strategic emphasis on Pharmaceutical / Bio-pharmaceutical and cutting-edge innovation. The company's manufacturing site is a cornerstone of its value chain, driving both large-scale biosimilar production and the development of R&D platforms critical to future growth.

Operating in a highly regulated environment—under standards such as ISO 9001:2015, cGMP, and international regulatory frameworks (FDA, EMA)—XXX (Firm Name) Limited recognizes that uninterrupted operations are essential to maintaining product integrity, regulatory compliance, and stakeholder trust.

This Business Continuity Plan (BCP) has been developed to safeguard the company's critical functions across all operational sites. It provides a robust framework to ensure rapid recovery and sustained performance during disruptive events, including internal system failures, natural disasters, pandemics, and regulatory or compliance challenges.

Key components of the BCP include:

Risk Identification – Proactive assessment of potential threats to operations and infrastructure.

Business Impact Analysis – Evaluation of critical processes and dependencies to prioritize recovery efforts.

Recovery Strategies – Tailored action plans to restore essential functions with minimal downtime.

Communication Protocols – Clear channels for internal and external communication during crises.

By integrating these elements into a cohesive preparedness strategy, XXX (Firm Name) Limited reinforces its commitment to resilience, regulatory excellence, and uninterrupted delivery of life-saving Pharmaceutical and Biopharmaceuticals.

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INTRODUCTION TO BUSINESS CONTINUITY PLAN (BCP):

A **Business Continuity Plan (BCP)** is a strategic framework designed to ensure that an organization can maintain or quickly resume its critical operations in the face of disruptive events—whether natural disasters, cyberattacks, pandemics, or system failures. It is a key component of **Business Continuity Management (BCM)**, which proactively prepares organizations to withstand and recover from crises with minimal impact on business functions. BCP goes beyond traditional disaster recovery by integrating:

Risk Assessment: Identifying potential threats and vulnerabilities that could impact operations.

Emergency Response: Establishing protocols for immediate action during incidents.

Crisis Communication: Ensuring timely and transparent communication with stakeholders.

Recovery Strategies: Defining procedures to restore essential services and infrastructure.

By safeguarding critical assets, data, personnel, and supply chains, a well-structured BCP helps organizations:

- ✔ Minimize operational downtime and financial losses
- ✔ Maintain compliance with industry regulations and standards
- ✔ Protect reputation and stakeholder trust
- ✔ Ensure uninterrupted delivery of products and services—especially vital in sectors like healthcare, finance, and public services

In essence, a Business Continuity Plan is not just a safety net—it's a strategic imperative that empowers organizations to navigate uncertainty with resilience and confidence.

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PURPOSE AND OBJECTIVES OF BUSINESS CONTINUITY PLAN (BCP):

The primary objective of this Business Continuity Plan (BCP) is to ensure the uninterrupted functioning of XXX (Firm Name) Limited' critical operations during and after disruptive events. This plan serves as a strategic safeguard to protect the organization's people, assets, and mission, while enabling swift recovery and sustained service delivery.

Key Objectives

The BCP is designed to:

Protect Life, Property, Environment, and Reputation.

Prioritize the safety of personnel, secure physical and intellectual assets, uphold environmental responsibilities, and preserve stakeholder trust.

Minimize Economic, Regulatory, and Quality-related Disruptions.

Reduce financial losses, maintain compliance with industry and governmental regulations, and ensure continued adherence to quality standards across all functions.

Enable Rapid and Structured Recovery of Operations and R&D.

Restore essential business and research activities through predefined recovery protocols, minimizing downtime and preserving innovation momentum.

Support Sustained Product Delivery to Customers.

Guarantee consistent supply of products and services to clients, especially those in healthcare and life sciences who depend on XXX (Firm Name) Limited' reliability.

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Scope and Guidelines

This BCP outlines a comprehensive framework for managing business interruptions, including:

Identification and Management of Failure Scenarios

Defines likely types of operational failures—such as IT outages, supply chain disruptions, facility damage, or personnel unavailability—and provides tailored response strategies for each.

Response Actions Based on Interruption Duration

Recommends specific actions depending on whether the disruption is short-term (minutes to hours), medium-term (days), or long-term (weeks or more), ensuring appropriate escalation and resource allocation.

Transition from Business Interruption to Disaster Recovery

Establishes clear criteria for recognizing when an incident escalates into a disaster-level event, triggering the activation of the Disaster Recovery Plan (DRP) and mobilizing cross-functional recovery teams.

SCOPE AND APPLICABILITY:

This Business Continuity Plan (BCP) is applicable across the entire operational landscape of XXX (Firm Name) Limited, encompassing all departments, functions, and stakeholders that contribute to the continuity of essential business activities. The plan ensures that every critical component of the organization is prepared to respond effectively to disruptions and resume operations with minimal delay.

Organizational Coverage

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The BCP applies to:

All Departments and Sites

Including—but not limited to—corporate offices, research facilities, manufacturing units, and satellite locations across XXX (Firm Name) Limited' operational footprint.

Associated Utility Systems

Vital infrastructure such as: Power Supply, HVAC (Heating, Ventilation, and Air Conditioning), and Water Systems, which support both operational and safety requirements.

Core Operational Functions

Quality Assurance (QA): Ensuring product integrity and regulatory compliance

Engineering: Maintaining equipment and facility readiness

Information Technology (IT): Safeguarding digital infrastructure and data continuity

Warehousing and Logistics: Managing inventory, distribution, and supply chain flow

Support and Administrative Functions

Procurement: Securing essential materials and services

Safety and EHS: Protecting personnel and environmental health

Distribution: Ensuring timely delivery of products to customers

Human Resources (HR): Supporting workforce continuity and communication

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Stakeholder Inclusion

The BCP also extends to:

Internal Stakeholders

Employees, leadership teams, and cross-functional units responsible for executing continuity and recovery protocols.

External Partners

Supply Chain Entities: Vendors, suppliers, and logistics providers critical to product flow and service delivery.

Contract Service Providers: Third-party collaborators involved in essential operations, including outsourced R&D, IT support, and facility management.

Integration with the Business Continuity Management (BCM) Framework:

This Business Continuity Plan (BCP) is a critical component of XXX (Firm Name) Limited's overarching Business Continuity Management (BCM) strategy. It does not operate in isolation but interfaces seamlessly with several specialized plans that collectively ensure organizational resilience, safety, and operational continuity during adverse events.

Disaster Recovery Plan (DRP)

The Disaster Recovery Plan is a technology-focused subset of the BCM framework, designed to:

Restore IT systems, network infrastructure, and digital platforms following cyber incidents, hardware failures, or data breaches.

Recover critical data and ensure integrity and availability of digital assets.

Minimize downtime of essential applications and research systems that support XXX (Firm Name) Limited's R&D and operational functions.

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Coordinate with IT teams and vendors to execute recovery protocols within defined Recovery Time Objectives (RTOs)

Crisis Communication Plan

The Crisis Communication Plan ensures transparent, timely, and effective communication during disruptive events. It includes:

Internal communication protocols for notifying employees, leadership, and response teams

External communication strategies for engaging with customers, suppliers, regulatory bodies, and the public

Guidelines for media handling, public disclosures, and regulatory liaison, ensuring compliance and reputation management

Designated spokespersons and escalation paths for message approval and dissemination

Emergency Response Plan (ERP)

The Emergency Response Plan addresses immediate physical threats and safety incidents, including:

Fire outbreaks, chemical spills, equipment malfunctions, and workplace accidents

Evacuation procedures, first-aid response, and coordination with local emergency services

Roles and responsibilities of emergency response teams, including Safety, EHS, and Facility Management

Post-incident reporting and corrective action protocols to prevent recurrence

Quality Risk Management Plan (QRM)

The Quality Risk Management Plan (QRM) is a vital component of XXX (Firm Name) Limited' Business Continuity Management (BCM) framework.

It provides a structured approach to identifying, evaluating, and mitigating risks that could impact Good Manufacturing Practices (GMP), product quality, and regulatory compliance. Specifically, the QRM addresses:

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GMP-related Deviations: Systematic assessment and resolution of deviations from established manufacturing protocols to prevent recurrence and ensure product integrity.

Corrective and Preventive Actions (CAPAs): Implementation of targeted actions to eliminate root causes of quality issues and proactively prevent future occurrences.

Regulatory Reporting: Timely and accurate communication with regulatory authorities regarding quality events, ensuring transparency and adherence to compliance obligations.

ORGANIZATION OVERVIEW:**COMPANY PROFILE:**

XXX (Firm Name) Limited is a pioneering, innovation-driven Pharmaceutical and Biopharmaceutical company dedicated to advancing global healthcare through the development and manufacturing of high-quality Pharmaceutical / Bio-Pharmaceutical. Founded with a clear vision to make healthcare more affordable and accessible, XXX (Firm Name) Limited combines scientific excellence with regulatory rigor to deliver impactful solutions for chronic and life-altering diseases.

Mission and Vision

At the heart of XXX (Firm Name) Limited lies a commitment to:

Enhancing global healthcare affordability through cost-effective biologics.

Delivering safe, effective, and accessible treatments for chronic conditions such as diabetes, autoimmune disorders, and other high-burden diseases.

Driving innovation in Pharmaceutical and Biopharmaceutical research and development to meet evolving patient needs and regulatory expectations.

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Core Competencies

XXX (Firm Name) Limited' strengths span the entire Pharmaceutical and Biopharmaceutical value chain, including:

GMP-Compliant Manufacturing: Robust infrastructure for large-scale production under stringent Good Manufacturing Practices (GMP), ensuring global regulatory compliance.

Team and Talent

The company's success is driven by a multidisciplinary team of:

Scientists and Technologists: Innovators in molecular biology, bioanalytic, and process engineering.

Regulatory Experts: Specialists in global regulatory frameworks, dossier preparation, and compliance strategy.

Quality and Operations Professionals: Champions of product integrity, safety, and supply chain excellence.

Global Outlook

With a strong foundation in R&D and manufacturing, XXX (Firm Name) Limited is strategically positioned to become a global leader in biosimilar solutions, serving diverse markets with high-quality, affordable biologics. The company actively collaborates with international partners, regulatory bodies, and healthcare providers to expand its reach and impact.

FACILITY OVERVIEW:

The manufacturing facility of XXX (Firm Name) Limited is a cutting-edge Pharmaceutical and Biopharmaceutical production site, purpose-built to support the development and large-scale manufacturing. Designed with precision and compliance at its core, the facility reflects XXX (Firm Name) Limited' commitment to quality, innovation, and global healthcare standards.

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Regulatory Compliance and Certification

Operates under ISO 9001:2015 certification, ensuring a robust quality management system across all operations

Fully aligned with Good Manufacturing Practice (GMP) guidelines issued by:

World Health Organization (WHO)

Central Drugs Standard Control Organization (CDSCO)

Other international regulatory authorities, enabling global market access and regulatory harmonization

Facility Infrastructure and Capabilities

The site integrates advanced bioprocessing technologies and infrastructure across the full production lifecycle:

Fermentation Suites (Upstream Processing)

Equipped with scalable bioreactors and controlled environments for high-yield microbial and mammalian cell culture

Chromatography and Filtration Units (Downstream Processing)

Incorporates high-performance purification systems to ensure product purity, potency, and consistency

Aseptic Fill-Finish Line (Grade A/B Environment)

Supports sterile formulation, filling, and packaging of drug products under stringent environmental controls to prevent contamination

Quality Control Laboratories

Chemical Testing: Assays for identity, potency, and impurity profiling

Microbiological Testing: Sterility, endotoxin, and bioburden assessments

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Analytical Testing: Advanced instrumentation for structural and functional characterization of biologics

Cold Storage and Warehousing

Temperature-controlled zones for raw materials, intermediates, and finished products, ensuring integrity throughout the supply chain

Engineering and Utility Infrastructure

Includes HVAC systems, purified water generation, clean steam, and backup power to support uninterrupted operations

Operational Flexibility

The facility is designed to accommodate both:

Commercial-Scale Production: High-volume manufacturing to meet global demand

Pilot-Scale R&D Operations: Agile development of new molecules, process optimization, and scale-up studies

Layout and Compliance Design

Unidirectional Flow of Materials and Personnel: Minimizes cross-contamination and supports cleanroom classification integrity

Contamination Control Measures: Includes airlocks, pressure differentials, and validated cleaning protocols

Regulatory-Ready Architecture: Facilitates inspections, audits, and documentation in line with global compliance expectations

PRODUCT PORTFOLIO:

XXX (Firm Name) Limited Product Portfolio and Development Focus

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XXX (Firm Name) Limited is at the forefront of Pharmaceutical and Biopharmaceutical innovation, with a strategic focus on the development and manufacturing of biosimilars that address high-burden chronic diseases. Leveraging advanced platforms and a robust regulatory framework, XXX (Firm Name) Limited is committed to delivering safe, effective, and affordable biologics to global markets.

ORGANIZATIONAL STRUCTURE AND KEY CONTACT:

XXX (Firm Name) Limited manufacturing facility operates under a robust and clearly defined organizational framework that ensures operational efficiency, regulatory compliance, and seamless cross-functional coordination. Each department is led by seasoned professionals who bring domain expertise and strategic oversight to their respective functions.

Site Head / Plant Head

Holds overall responsibility for the plant's performance, safety, and compliance

Oversees strategic planning, resource allocation, and cross-functional alignment

Acts as the primary liaison with corporate leadership, regulatory bodies, and external stakeholders

Ensures adherence to GMP, EHS standards, and business continuity protocols

Head of Production

Leads all manufacturing operations, including upstream and downstream processes

Ensures process optimization, batch consistency, and production scheduling

Coordinates with QA, QC, and Engineering to maintain uninterrupted operations

Implements lean manufacturing practices and continuous improvement initiatives

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Head of Quality Assurance (QA)

Responsible for Good Manufacturing Practice (GMP) compliance across all operations

Oversees documentation control, standard operating procedures (SOPs), and batch release protocols

Manages internal audits, regulatory inspections, and quality risk management

Ensures that all products meet regulatory and pharmacopeial standards

Head of Quality Control (QC)

Manages analytical testing of raw materials, in-process samples, and finished products

Oversees stability studies, method validation, and environmental monitoring

Maintains compliance with pharmacopeial requirements and data integrity standards

Collaborates with QA to resolve deviations and ensure timely batch disposition

Head of Research & Development (R&D)

Drives new product development, formulation innovation, and biosimilar characterization

Leads technology transfer from lab-scale to commercial manufacturing

Coordinates with regulatory affairs for clinical trial support and dossier preparation

Fosters a culture of scientific excellence and intellectual property generation

Engineering and Maintenance Lead

Oversees plant utilities, HVAC systems, water systems, and critical equipment

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Ensures preventive and corrective maintenance to minimize downtime

Supports qualification and validation of equipment and infrastructure

Plays a key role in energy efficiency, sustainability, and facility upgrades

IT and Digital Systems Lead

Manages IT infrastructure, network security, and data backup protocols

Ensures cybersecurity compliance, access control, and disaster recovery readiness

Supports digital transformation initiatives, including MES, LIMS, and ERP systems

Facilitates data integrity, electronic batch records, and regulatory audit readiness

GOVERNANCE AND POLICY FRAMEWORK:**BCP POLICY STATEMENT:**

At XXX (Firm Name) Limited, resilience is a strategic imperative. We are deeply committed to ensuring the uninterrupted operation of our critical functions—even in the face of unforeseen disruptions, whether internal (e.g., system failures, personnel shortages) or external (e.g., natural disasters, geopolitical events, pandemics).

Our Business Continuity Plan (BCP) is a cornerstone of our risk management framework, designed to provide a structured, proactive, and responsive approach to safeguarding our operations, assets, and stakeholder interests.

Key Objectives of the BCP Policy

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Risk Identification & Assessment: Systematic evaluation of potential threats to operations, including supply chain vulnerabilities, IT infrastructure risks, and regulatory disruptions.

Impact Analysis: Detailed Business Impact Analysis (BIA) to prioritize functions based on criticality and recovery time objectives (RTOs).

Preventive Strategies: Implementation of safeguards such as redundancy in utilities, secure data backups, alternate suppliers, and workforce cross-training.

Incident Response Protocols: Clearly defined escalation paths, communication plans, and decision-making hierarchies to ensure swift and coordinated action.

Recovery & Restoration: Stepwise recovery procedures to restore operations to normal or near-normal levels within defined timelines.

Compliance & Alignment: Adherence to global standards such as ISO 22301, GxP guidelines, and customer-specific continuity requirements.

Departmental Integration

Our BCM policy is **embedded across all functional areas**, ensuring that each department contributes to and benefits from a unified continuity strategy:

Production: Backup utilities, alternate manufacturing lines, and inventory buffers.

Quality Assurance & Control: Secure storage of documentation, validated alternate testing sites.

R&D: Protection of intellectual property and continuity of critical research programs.

Engineering & Maintenance: Redundant systems and emergency response protocols.

IT & Digital Systems: Cybersecurity measures, cloud-based backups, and disaster recovery plans.

Corporate & Regulatory Affairs: Stakeholder communication, regulatory reporting, and compliance continuity.

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Continuous Improvement & Testing

Regular BCP drills, tabletop exercises, and scenario simulations to validate readiness

Periodic policy reviews and gap assessments to incorporate lessons learned and evolving risks

Engagement with external experts and auditors to benchmark against industry best practices

ROLES AND RESPONSIBILITIES:

The successful execution of XXX (Firm Name) Limited Business Continuity Plan (BCP) depends on a clearly defined governance structure, with roles and responsibilities distributed across all levels of the organization. This ensures coordinated action, accountability, and sustained resilience during disruptive events.

Executive Management

Strategic Oversight: Approves the overall BCP framework and ensures alignment with corporate objectives, regulatory requirements, and risk appetite

Resource Allocation: Commits financial, technological, and human resources necessary for continuity planning, testing, and execution

Policy Endorsement: Champions a culture of preparedness and resilience across the organization

Regulatory Interface: Engages with external stakeholders and regulators during major incidents or audits

BCP Coordinator

Plan Ownership: Maintains the BCP document, ensuring it remains current, relevant, and reflective of organizational changes

Testing & Validation: Organizes periodic drills, tabletop exercises, and scenario simulations to assess readiness

Training & Awareness: Facilitates employee education programs and ensures departmental understanding of continuity protocols

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Audit & Review: Coordinates internal reviews and external audits to validate plan effectiveness and identify improvement areas

Incident Response Team (IRT)

Operational Execution: Leads the tactical response during business interruptions, coordinating recovery efforts across functions

Situation Assessment: Evaluates the nature and impact of the disruption, activates appropriate response protocols

Communication Hub: Interfaces with the Crisis Communication Team to ensure timely and accurate stakeholder updates

Post-Incident Review: Conducts debrief and contributes to root cause analysis and corrective actions

Department Heads

Functional Continuity Planning: Develops and maintains department-specific continuity strategies, aligned with the central BCP

Team Readiness: Ensures that staff are trained, aware of their roles, and equipped to respond effectively

Resource Management: Identifies critical assets, alternate workflows, and recovery priorities within their domain

Reporting & Escalation: Communicates status updates and escalates issues to the BCP Coordinator and IRT as needed

All Employees

Awareness & Participation: Understand their individual roles during disruptions and actively participate in training sessions

Compliance: Follow established protocols during incidents, including evacuation, communication, and recovery procedures

Feedback & Improvement: Contribute insights from drills and real events to enhance future preparedness

Responsibility: Maintain vigilance and report potential risks or vulnerabilities that may impact continuity

BCP COMMITTEE STRUCTURE:

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To ensure the effective governance and execution of its Business Continuity Plan (BCP), XXX (Firm Name) Limited has established a dedicated BCP Committee at all operational sites. This committee plays a pivotal role in overseeing the planning, implementation, monitoring, and continuous enhancement of business continuity strategies across the organization.

Purpose and Mandate

The BCP Committee is entrusted with:

Strategic Oversight of continuity planning and risk mitigation initiatives

Cross-functional Coordination to ensure alignment of departmental continuity strategies

Periodic Review and Testing of BCP protocols to validate readiness and identify improvement areas

Post-Incident Evaluation to assess response effectiveness and implement corrective actions

Regulatory and Audit Preparedness to ensure documentation and practices meet global compliance standards

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Committee Composition

The committee comprises senior representatives from key operational and support functions, ensuring holistic coverage of all critical business areas:

Department	Role in BCP Committee
Quality Assurance (QA)	Ensures GMP compliance, documentation integrity, and quality continuity during disruptions
Production	Oversees manufacturing continuity, alternate batch planning, and resource reallocation
Engineering	Manages utility resilience, equipment readiness, and infrastructure recovery
Regulatory Affairs	Coordinates regulatory communication, reporting obligations, and compliance alignment
Research & Development (R&D)	Safeguards ongoing research, tech transfer continuity, and IP protection
IT and Data Integrity	Maintains digital infrastructure, cybersecurity, and disaster recovery protocols
Human Resources (HR)	Supports workforce continuity, communication, and emergency staffing plans
Warehouse and Logistics	Ensures material availability, inventory control, and supply chain resilience

Meeting Frequency**Quarterly Meetings:**

Review BCP status, risk landscape, and departmental updates

Plan and schedule drills, audits, and training programs

Discuss regulatory changes and their impact on continuity planning

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Post-Incident Reviews:

Conduct root cause analysis and evaluate response effectiveness

Document lessons learned and update BCP protocols accordingly

Recommend corrective and preventive actions (CAPAs)

Regulatory Alignment and Standards Integration in XXX (Firm Name) Limited' BCP Program

XXX (Firm Name) Limited operates within a highly regulated Pharmaceutical and Biopharmaceutical environment, where uninterrupted operations and product integrity are paramount. To uphold these commitments, the organization's Business Continuity Planning (BCP) program is meticulously aligned with globally recognized standards and regulatory frameworks. This alignment ensures that continuity strategies are not only robust and risk-responsive but also compliant with industry expectations and patient-centric obligations.

Standards Incorporated into the BCP Framework

The BCP program draws upon the following foundational standards:

Good Manufacturing Practices (GMP)

Ensures that products are consistently manufactured and controlled according to stringent quality standards

Supports continuity of critical quality systems during disruptions

Reinforces compliance with regulatory authorities such as FDA, EMA, WHO, and CDSCO

ISO 9001:2015 – Quality Management System (QMS)

Establishes a structured approach to quality governance and operational excellence

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Enhances resilience through risk-based thinking, process standardization, and continual improvement

Facilitates cross-functional accountability and documentation integrity during continuity events

ISO 14001:2015 – Environmental Management System (EMS)

Promotes sustainable operations and environmental risk mitigation

Ensures continuity of environmental controls and compliance during emergencies

Aligns with XXX (Firm Name) Limited' commitment to responsible manufacturing and ecological stewardship

ISO 22301:2019 – Business Continuity Management System (BCMS)

Provides a comprehensive framework for identifying threats, assessing impacts, and developing recovery strategies

Enables structured planning, testing, and improvement of continuity protocols

Supports stakeholder confidence, regulatory readiness, and uninterrupted product availability

Strategic Outcomes:

By integrating these standards into its BCP architecture, XXX (Firm Name) Limited achieves:

Regulatory Assurance: Demonstrates preparedness and compliance across global jurisdictions

Operational Resilience: Maintains core functions and product supply during adverse events

Risk Mitigation: Proactively identifies vulnerabilities and implements safeguards

Patient Commitment: Ensures consistent access to high-quality biosimilar therapies, even during disruptions

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ABBREVIATIONS:

Abbreviation	Full Form	Abbreviation	Full Form
BCP	Business Continuity Plan	MCB/WCB	Master Cell Bank / Working Cell Bank
BIA	Business Impact Analysis	MTD	Maximum Tolerable Downtime
CBF	Critical Business Function	OOS	Out of Specification
CAPA	Corrective and Preventive Action	OOT	Out of Trend
CMC	Chemistry, Manufacturing, and Controls	PLC	Programmable Logic Controller
CTD	Common Technical Document	QA	Quality Assurance
CFT	Cross-Functional Team	QC	Quality Control
CIP	Clean-in-Place	QMS	Quality Management System
SIP	Sterilize-in-Place	RPO	Recovery Point Objective
ERT	Emergency Response Team	RTO	Recovery Time Objective
GMP	Good Manufacturing Practice	SCADA	Supervisory Control and Data Acquisition
HVAC	Heating, Ventilation, and Air Conditioning	SOP	Standard Operating Procedure
IT	Information Technology	TOC	Total Organic Carbon
LIMS	Laboratory Information Management System	WFI	Water for Injection

DEFINITIONS:

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Term	Definition		
Business Continuity Plan (BCP)	A documented framework outlining procedures and instructions to ensure critical business functions can continue during and after a disruption.		
Business Impact Analysis (BIA)	A systematic process to identify critical business functions, assess the impact of disruptions, and establish recovery priorities and timelines.		
Critical Business Function (CBF)	A process or activity essential to the delivery of key products or services, whose interruption would significantly impact operations.		
Corrective and Preventive Action (CAPA)	Actions taken to eliminate the causes of nonconformities or other undesirable situations to prevent recurrence.		
Recovery Time Objective (RTO)	The maximum acceptable time allowed to restore a business process or system after a disruption.		
Recovery Point Objective (RPO)	The maximum acceptable amount of data loss measured in time that an organization can tolerate during a disruption.		
Maximum Tolerable Downtime (MTD)	The total period that a business process can be inoperable without causing irreparable harm to the organization.		

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Term	Definition		
Deviation	A departure from approved procedures, standards, or specifications that requires documentation, investigation, and resolution.		
Good Manufacturing Practice (GMP)	Regulatory standards that ensure products are consistently produced and controlled according to quality standards.		
Clean-in-Place (CIP)	An automated cleaning process for the interior surfaces of equipment without disassembly.		
Sterilize-in-Place (SIP)	A procedure to sterilize the internal surfaces of process equipment using steam or other sterilant without disassembly.		
Out of Specification (OOS)	A test result that falls outside pre-established acceptance criteria.		
Out of Trend (OOT)	A result that, while within specification, shows an unexpected or abnormal pattern compared to historical data.		
Emergency Response Team (ERT)	A trained group of personnel responsible for managing and mitigating emergency situations.		
Cross-Functional Team (CFT)	A group of individuals from different functional areas working together toward a common goal.		

REFERENCES:

ISO 22301:2019 – Security and resilience – Business continuity management systems – Requirements.

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ICH Q9 (R1) – Quality Risk Management

EU GMP Guidelines – EudraLex Volume 4, Part I & II

WHO Technical Report Series – Annexes on Good Manufacturing Practices

XXX (Firm Name) Limited Internal SOPs on Business Continuity, Deviation Management, and Emergency Response

U.S. FDA 21 CFR Parts 210 and 211 – Current Good Manufacturing Practice for Finished Pharmaceuticals

ISO 9001:2015 – Quality Management Systems – Requirements

Business Impact Analysis (BIA) Methodology – XXX (Firm Name) Limited

The Business Impact Analysis (BIA) is a foundational element of XXX (Firm Name) Limited Business Continuity Planning (BCP) framework. It provides a structured, data-driven approach to identifying critical business functions (CBFs), evaluating potential disruption scenarios, and prioritizing recovery strategies. The methodology is aligned with ISO 22301:2019 and regulatory expectations from WHO-GMP, FDA, and EMA, ensuring both operational resilience and compliance integrity.

Step 1: Identification of Critical Business Functions (CBFs)

Each department is systematically assessed to identify processes essential for maintaining:

- Operational continuity
- Regulatory compliance
- Product quality and safety
- Customer commitments

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Departments Assessed:

Production (Upstream, Downstream, Fill-Finish)

Quality Assurance (QA)

Quality Control (QC)

Microbiology

Research & Development (R&D)

Regulatory Affairs

IT & Data Integrity

Engineering & Utilities

Warehouse & Logistics

Human Resources & Administration

Safety & EHS

Examples of CBFs:

Batch release and QMS oversight

Analytical and microbiological testing

Inventory and cold chain management

Data backup and digital infrastructure

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Utility systems (HVAC, FW/WFI, DG sets)

Step 2: Scenario-Based Risk Assessment

Department-specific risk scenarios are evaluated to determine the severity and likelihood of disruptions. This includes:

Department	Example Risk Scenarios
Engineering	HVAC failure, boiler outage, briquette shortage
Production	Upstream/downstream process failure, aseptic fill-finish disruption
QA/QC	Batch release delays, analytical equipment failure
Microbiology	Environmental monitoring lapses, sterility test failures
Regulatory Affairs	Submission delays, data integrity issues
R&D	Tech transfer delays, process development bottlenecks
IT	DMS/LMS failure, cybersecurity breach, data loss
Warehouse	Inventory mismanagement, cold chain failure
HR/Admin	Workforce unavailability, safety incident
Safety & EHS	Fire, chemical spill, environmental non-compliance

Step 3: Impact Categorization

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Each CBF is analysed across four impact domains:

Financial Impact: Revenue loss, increased operational costs, contractual penalties

Business Disruption: Workflow delays, production halts, service interruptions

Compliance Impact: Regulatory deviations, audit failures, data integrity risks

Reputational Impact: Customer dissatisfaction, brand erosion, public trust loss

This categorization helps prioritize recovery efforts based on severity and strategic importance.

Step 4: Recovery Objective Definition

For each CBF, the following recovery metrics are defined:

Recovery Time Objective (RTO): Maximum allowable time to restore a function before significant impact

Recovery Point Objective (RPO): Maximum acceptable data loss measured in time (e.g., last backup timestamp)

Maximum Tolerable Downtime (MTD): Longest duration a function can remain inoperative before causing irreversible damage

These metrics are determined based on:

Functional criticality

Interdependencies across departments

Historical disruption data

Regulatory and customer expectations

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IDENTIFICATION OF CRITICAL BUSINESS FUNCTIONS (CBF):

Refer Annexure I – Risk Assessment & Impact Analysis and Annexure II – Critical Business Functions Matrix for the identification of department-wise Critical Business Functions (CBFs) essential to ensure continuity of operations, regulatory compliance, and customer commitments at XXX (Firm Name) Limited.

Scenario-Based Risk Assessment

This section presents a structured, high-level evaluation of critical threat scenarios that could significantly disrupt operations at XXX (Firm Name) Limited' manufacturing and research site. Each scenario is assessed for its potential impact across five key operational domains—personnel, processes, equipment, supply chain, and IT infrastructure—ensuring a holistic understanding of organizational vulnerabilities and resilience.

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HVAC Risk Management Plan – Engineering Perspective

Scenario	Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Loss of temperature, humidity, or pressure control in cleanrooms	Batch deviation, microbial contamination, regulatory non-compliance	Medium	High	High	Manufacturing, QA, Engineering	Periodic preventive maintenance (PPM), calibration of sensors (Temp, RH, Pressure), BMS monitoring	Activate Emergency Response SOP, QA to assess impact on in-process stored materials, document deviation
Failure of HEPA filters or AHU fans in Grade A/B/C zones	Airborne contamination, product rejection, audit observations	Medium	High	High	Manufacturing, QA, Engineering	Scheduled DOP/PAO testing; timely HEPA replacement, AHU performance checks	Halt batch processing, isolate and label affected materials; QA to evaluate and clear
HVAC shutdown leading to microbial/particulate excursion	Environmental excursion, risk to sterility assurance, production delays	Low	High	Medium	QA, Engineering, Production	Continuous EM and particle monitoring, BMS alerts; UPS for critical systems	Notify QA/Engineering, sanitize affected areas, perform EM before resumption

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Pressure cascade failure causing reverse airflow or ingress from unclassified zones	Cross-contamination, deviation from cleanroom classification	Low	High	Medium	Engineering, QA, Production	BMS-configured pressure interlocks and alarms; periodic pressure mapping	Seal affected zones; conduct root cause analysis; QA to approve restart
Cold/hot spots in storage areas	Stability data deviation, product degradation, regulatory risk	Medium	Medium	Medium	QA, Stability, Warehouse	Quarterly HVAC airflow mapping; corrective balancing; temperature log audits	Transfer products to validated alternate locations; QA and stability team to assess
HVAC downtime in critical production blocks	Production halt, batch delays, deviation reporting	Medium	Medium	Medium	Production, Engineering, QA	AMC with OEM/vendor; maintain critical spares (belts, motors, filters); redundancy planning	Initiate recovery post-rectification; document deviation; QA to validate restart

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
HVAC malfunction during facility equipment qualification	Delay in commissioning, requalification costs, project timeline impact	Medium	Medium	Medium	Validation, QA, Engineering	Maintain qualification documentation, align revalidation schedule with HVAC readiness	Reschedule qualification; revise impact assessment; QA Validation to approve
Uncontrolled HVAC restart post-maintenance	Pressure imbalance, contamination risk, equipment stress	Low	High	Medium	Engineering, QA	SOP for controlled restart; validation of airflow and pressure post-maintenance	QA to verify environmental parameters; Engineering to log restart data

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Water Shortage Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Unavailability of DM, PW, or WFI	Disruption of production processes, cleaning, and sterilization; risk of batch delays and non-compliance	Medium	High	High	Production, QA, Engineering	Maintain adequate RO, DM, PW, and WFI tank capacities based on peak daily usage; monitor consumption trends	Activate standby borewell pump; notify Engineering and QA; initiate deviation documentation if impact exceeds threshold
Delay in batch manufacturing and cleaning activities	Extended turnaround times, reduced throughput, potential backlog in production schedule	Medium	Medium	Medium	Production, QA, Planning	Install and regularly service standby borewell pump; maintain vendor contracts for emergency water supply	Trigger water tanker procurement via approved vendor; validate water quality before use; inform Planning for rescheduling

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Sterile manufacturing disruption due to WFI shortage	Risk to sterility assurance, halted aseptic operations, potential batch rejection	Low	High	Medium	Sterile Production, QA, Engineering	Schedule and track WFI usage aligned with production planning, maintain buffer stock	Prioritize WFI allocation for sterile blocks, QA to assess impact, escalate if downtime exceeds 3 hours
Water insufficiency affecting product quality and compliance	Deviations in cleaning validation, equipment performance, and environmental control	Medium	High	High	QA, Production, Engineering	Quarterly preventive maintenance of water systems and pumps, real-time monitoring of supply pressure and flow	Inform QA, halt critical processes if supply drops below validated thresholds, initiate CAPA if product impact is suspected
Manual top-up delays for DM/PW/WFI tanks	Operational inefficiencies, risk of tank depletion	Medium	Medium	Medium	Engineering, Production	Automate level sensing and top-up systems, configure alerts for low-level thresholds	Engineering to monitor and refill tanks promptly, log response time and corrective actions

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
	increased manpower dependency						
Unnoticed depletion due to manual monitoring	Sudden operational halt, risk of contamination due to low flow or pressure	Medium	High	High	Utility, QA, Production	Display real-time tank level dashboard in utility control room; integrate with BMS alerts	Maintain continuous communication among QA, Production, and Utility teams; initiate emergency water conservation protocol
No formal prioritization of water allocation during shortage	Inefficient resource use, delayed recovery, potential impact on critical operations	Medium	Medium	Medium	Site Leadership, QA, Utility	Develop Water Shortage SOP with consumption hierarchy: WFI > PW > DM > RO > Utility	Escalate to Emergency Response Team (ERT) and Site Head if recovery exceeds 3 hours; activate prioritization matrix

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Power Outage Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Power outage during production or batch processing	Sudden equipment shutdown, batch failure, data loss, deviation from validated parameters	Medium	High	High	Production, QA, Engineering	Maintain DG sets with auto-start synchronization; validate capacity against peak load	Initiate emergency power shift via DG; monitor load transfer; QA to assess batch impact
DG set failure during main power loss	HVAC, AHU, cleanroom pressure cascade failure; halted sterile operations	Low	High	Medium	Engineering, QA, Sterile Production	Daily DG start-up trials; weekly full-load test runs; alarm integration with BMS	Activate Emergency Response SOP; inform QA and Engineering; initiate deviation documentation
Loss of WFI generation, HVAC,	Sterility assurance risk; temperature	Low	High	Medium	QA, Engineering,	Maintain diesel inventory for 8–12 hours; link	Use portable UPS for priority systems;

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
AHU, cold storage due to DG failure	excursions, halted aseptic processing				Sterile Production	critical systems to UPS-DG circuits	transfer critical loads; QA to validate restart
Breakdown of DG alternator, radiator, or control panel during extended use	Extended outage, inability to sustain operations, fire risk	Medium	Medium	Medium	Engineering, EHS	OEM-based preventive maintenance, AMC coverage, spare part inventory	Shift to alternate DG (if available); engage AMC vendor for emergency repair
Failure in cold chain support (e.g., stability chambers, cold rooms)	Product degradation, excursion in stability data, regulatory non-compliance	Medium	High	High	QA, Stability, Warehouse	Temperature alarms, cold storage mapped to DG supply, backup chillers	Transfer products to alternate qualified storage, notify QA; document excursion and CAPA
Fire or safety hazard due to fuel leakage or overheating	Personnel safety risk, equipment damage, regulatory violation	Low	High	Medium	EHS, Engineering	Fire extinguishers, spill kits, flameproof lighting	Isolate DG set; shut down fuel line, initiate

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
						weekly housekeeping audits	ERT firefighting SOP, log incident
Delay in QA or regulatory operations due to server/SCADA shutdown	Data integrity risk, delayed batch release, audit observations	Medium	Medium	Medium	QA, IT, Regulatory Affairs	Link QA/QMS/SCADA servers to online UPS + DG, validate backup protocols	Ensure IT backup is active; document and justify data delays; inform regulatory team if needed

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Boiler Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Steam supply interruption	Disruption of sterilization, HVAC, and manufacturing processes; delayed batch execution; deviation from validated cycles	Medium	High	High	Production, QA, Engineering	Schedule preventive maintenance and calibration of boiler, sensors, and control systems; monitor steam pressure trends	Inform QA and Production; halt dependent operations; shift to alternate batch steps if feasible; document deviation
Boiler overpressure or rupture	Equipment damage; personnel injury; fire hazard; regulatory non-compliance	Low	High	Medium	Engineering, EHS, QA	Ensure pressure relief valves, alarms, and interlocks are functional and tested; conduct weekly safety checks	Activate Emergency Response SOP; isolate boiler via main steam and fuel shut-off; engage ERT; initiate incident investigation

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Poor steam quality (wet steam, non-condensable gases)	Incomplete sterilization, equipment malfunction, compromised product quality	Medium	Medium	Medium	QA, Production, Engineering	Maintain boiler water quality per SOP; perform regular blowdown; monitor feedwater treatment and condensate return	Notify QA and Engineering; assess impact on ongoing sterilization cycles; initiate revalidation if required
Fuel leakage or combustion failure	Fire/explosion risk, operational shutdown, safety hazard	Low	High	Medium	EHS, Engineering	Daily inspection of fuel lines, flame sensors, and burner assembly; maintain fire extinguishers and spill kits	Evacuate boiler area; isolate fuel source; engage ERT; initiate firefighting protocol; log incident and corrective actions

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Expired certification or missing log records	Audit non-compliance, regulatory penalties, operational suspension	Medium	Medium	Medium	QA, Engineering, Safety	Maintain statutory certifications, update logbooks for pressure, hours, fuel usage, inspections, conduct internal audits	Escalate to Safety Officer and Engineering Head, resolve documentation gaps; prepare for regulatory inspection
Electrical control panel or automation failure	Boiler shutdown, delayed steam supply, manual override risks	Medium	Medium	Medium	Engineering, QA	Enable backup panel or manual control mode; periodic testing of control systems and PLCs	Switch to manual mode if safe, log deviation; initiate corrective action and root cause analysis
Noise, vibration, or structural stress	Indication of imbalance, risk of mechanical failure or rupture	Medium	Medium	Medium	Engineering, EHS	Routine vibration analysis; mechanical	Shut down boiler; inspect for imbalance, cracks, or wear; resume only after

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
						inspection of mountings, supports, and insulation	mechanical clearance and QA approval

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Briquettes Shortage Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Inability to generate steam for sterilization, HVAC, cleaning, and process operations	Halted batch processing, compromised cleanroom conditions, delayed SIP/CIP cycles	Medium	High	High	Production, QA, Engineering	Maintain 5–7 days of briquette buffer stock; monitor daily consumption trends	Immediately alert Purchase and Vendor Management; activate emergency procurement protocol
Boiler downtime causing delay in batch processing	Mixed production timelines, backlog in manufacturing schedule, deviation from planned output	Medium	Medium	Medium	Production, Planning, QA	Forecast briquette consumption based on production planning and seasonal demand	Prioritize briquette usage for critical utilities (WFI generation, sterilizers); reschedule non-critical batches
Impact on product quality due to loss	Risk to sterility assurance, temperature-humidity	Low	High	Medium	QA, Production, Engineering	Qualify minimum two alternate briquette vendors	Notify QA and Production; assess impact; reschedule

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
of environmental control	excursions, potential batch rejection					under QA oversight; validate supply quality	non-critical activities; document deviation
Reduced efficiency in heating operations (e.g., fermenter jackets, hot water lines)	Suboptimal process conditions, reduced yield, equipment stress	Medium	Medium	Medium	Engineering, Production	Monthly reconciliation of briquette stock vs. usage logs; monitor boiler performance	Coordinate with Admin/Procurement for expedited dispatch from backup supplier; log response time
Regulatory non-compliance due to missed cleaning/SIP timelines	Audit observations, deviation reports, potential regulatory penalties	Low	High	Medium	QA, Regulatory Affairs, Production	Maintain supply contracts with minimum commitment and defined lead time;	Log deviation; obtain QA approval for rescheduling; update cleaning/SIP records and justification

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
						track vendor reliability	
Financial loss from last-minute procurement at inflated rates	Budget overruns, strained vendor relationships, emergency logistics costs	Medium	Medium	Medium	Finance, Procurement, Utility	Integrate briquette buffer planning into annual utility budgeting; monitor market trends	Escalate to Site Head if supply delay exceeds 48 hours; initiate cost-impact assessment and vendor negotiation

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Cooling Tower Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Reduced cooling efficiency leading to overheating of HVAC chillers, compressors, fermenters	Equipment shutdown, temperature excursions in cleanrooms, halted fermentation or HVAC operations	Medium	High	High	Engineering, Production, QA	Regular cleaning of drift eliminators, fill packs, and nozzles; verify water flow rate and fan operation	Stop non-critical cooling loads; prioritize cooling for essential equipment; monitor temperature recovery
Microbial contamination (e.g., <i>Legionella</i>) in cooling water	Risk to cleanroom air handling, health hazard, regulatory non-compliance	Medium	High	High	EHS, QA, Engineering	Routine dosing with biocides and anti-scale agents; maintain blowdown schedules; monitor microbial counts	Isolate affected loop; notify EHS and QA; increase dosing; initiate disinfection protocol; document excursion
Cooling tower motor/fan failure	Production downtime, loss of cooling	Medium	Medium	Medium	Engineering, Production	Monthly preventive maintenance of motor,	Switch to standby cooling tower (if available);

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
	capacity, deviation from validated conditions					bearings, gearbox, maintain spare motor inventory	engage vendor for urgent repair, log downtime and recovery
Reduced efficiency due to high ambient temperature or scaling in heat exchangers	Inadequate cooling, temperature deviation in process equipment, reduced yield	High	Medium	High	Engineering, QA, Production	Monitor approach temperature; quarterly cleaning of heat exchangers; scale inhibitor dosing	Divert load to alternate chilling equipment, inform QA/Engineering if temperature exceeds limits; assess impact
Overflow or water spillage from cooling tower basin	Slip hazard, water loss, electrical risk, contamination of adjacent areas	Medium	Medium	Medium	EHS, Engineering	Install level controllers and overflow drains; inspect basin and piping for leaks	Isolate power, cordon off area; investigate root cause; repair and resume post EHS clearance

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Chemical handling incident during manual dosing	Operator injury, chemical spill, environmental hazard, audit finding	Low	High	Medium	EHS, Utility, Engineering	Use automated dosing pumps with containment bonds; PPE and training for operators; MSDS availability	Provide first aid; isolate leak; inform EHS; initiate spill control SOP; log incident and corrective actions
Structural damage or corrosion in tower body or support frame	Tower collapse risk, operational shutdown, safety hazard	Low	High	Medium	Engineering, EHS	Annual structural audit; corrosion control painting; maintain FRP integrity; inspect anchor points	Decommission damaged tower; shift load to alternate unit; initiate civil/mechanical repairs; QA to validate restart

Chiller Risk Management Plan

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Temperature control failure in cleanrooms, AHUs, cold rooms, or stability chambers	Environmental excursions, compromised product stability, deviation from validated conditions	Medium	High	High	QA, Engineering, Production, Stability	Scheduled preventive maintenance of compressors, condensers, evaporators, and sensors; monitor temperature trends	Notify QA and Engineering; shift temperature-sensitive materials to validated alternate storage; initiate deviation documentation
Chiller breakdown during fermentation, coating, or granulation	Process interruption, reduced yield, batch delays, equipment stress	Medium	Medium	Medium	Production, Engineering, QA	Maintain validated alternate chilling systems (e.g., backup chiller, buffer tank); test switchover protocols	Activate standby chiller or reroute chilled water from alternate loop; QA to assess process impact
Product degradation due to high temperature in raw	Loss of potency, stability data	Low	High	Medium	QA, Warehouse, Stability	Calibrate temperature sensors; maintain chiller	Transfer affected material to backup qualified area; document deviation; QA

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
material or finished goods storage	deviation, regulatory non-compliance					operation logs; conduct periodic audits	to assess impact and initiate CAPA
Alarm fatigue or late detection due to sensor/control panel error	Delayed response to excursions, risk of unnoticed failures, audit findings	Medium	Medium	Medium	Engineering, QA, IT	Connect chiller sensors to SCADA/BMS with real-time alarm monitoring; validate alert thresholds	Manual inspection if alert is triggered; escalate to ERT; initiate fault diagnosis and corrective action
Microbial growth in HVAC ducts due to prolonged chiller downtime	Contamination risk, deviation from cleanroom classification, sterility assurance impact	Low	High	Medium	QA, Engineering, EHS	Maintain positive pressure; regular duct sanitization; monitor air changes and microbial counts	Isolate affected zone; increase air changes using mobile AHUs; perform EM and sanitization before restart

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Chiller trip due to voltage fluctuation or overload	Sudden shutdown, temperature excursions, equipment damage	Medium	Medium	Medium	Engineering, QA, Utility	Install voltage stabilizers and surge protection; monitor ampere loads; conduct electrical audits	Reset breakers after safety check; log event; monitor closely during restart; QA to validate environmental recovery
Delay in restarting chiller affecting dependent systems	Extended downtime, cascading impact on HVAC, cold storage, and cleanroom conditions	Medium	Medium	Medium	Engineering, QA, Production	Train engineering staff on emergency chiller start-up protocols; maintain SOPs and restart checklists	Follow SOP for manual restart; inform all affected departments and QA; document deviation and recovery timeline

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Production Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Contamination in media preparation or fermentation vessels	Batch rejection, microbial contamination, regulatory deviation, loss of production time	Medium	High	High	Production, QA, Microbiology, Engineering	Validate CIP/SIP cycles, perform routine EM of vessels, transfer lines, and laminar hoods; use sterile connections	Halt process immediately; isolate contaminated batch; notify QA; initiate deviation investigation and CAPA
Cell growth inhibition or delayed fermentation due to media composition or pH deviation	Reduced biomass yield, extended fermentation time, downstream bottlenecks	Medium	Medium	Medium	Production, QA, QC, Formulation	Use qualified raw materials; validate media prep protocols; calibrate pH and DO sensors; monitor media lot history	Conduct RCA; prepare fresh media; reinstitute inoculum development if viable; document deviation

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Fermenter temperature, pH, or DO control failure	Suboptimal culture growth, altered metabolite profile, batch failure	Low	High	Medium	Production, Engineering, QA	SCADA-based monitoring with alarm thresholds; periodic calibration of probes and sensors	Shift to backup fermenter if available; adjust parameters; monitor viability curve; QA to assess salvageability
Power failure impacting fermenter controls (RPM, airflow, heating)	Process interruption, temperature excursions, cell viability loss	Medium	High	High	Engineering, Production, QA	Link fermenters to UPS and DG backup; test failover systems quarterly; monitor load capacity	Engage Engineering to restore power, stabilize parameters; assess impact on culture; document deviation
Equipment malfunction (agitator, seal, aeration failure)	Incomplete mixing, oxygen limitation, contamination risk	Medium	Medium	Medium	Engineering, Production, QA	Preventive maintenance schedule; maintain critical spares inventory	Shift batch if salvageable; log deviation; initiate

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
						(motors, seals, valves, pumps)	mechanical repair under QA supervision; reassess batch quality
Operator error during inoculation, sampling, or feeding	GMP deviation, contamination risk, batch rejection	Medium	Medium	Medium	Production, QA, Training	Conduct operator training and requalification; enforce SOP adherence; maintain logbook traceability	Isolate batch if deviation observed; QA to assess for reprocessing or rejection; initiate retraining if needed
Seed failure or non-growth due to inactive inoculum or improper transfer	Process delay, batch failure, loss of production slot	Low	High	Medium	Production, QA, Microbiology	Monitor OD, viable count, and morphology at each seed stage; maintain backup seed cultures	Redevelop seed from previous stage or master cell bank; document event and impact on timeline

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Delay in fermentation start impacting downstream timelines	Scheduling conflicts, reduced equipment utilization, QA release delays	Medium	Low	Low	Production, QA, Planning	Integrate upstream planning with downstream buffers; maintain flexible scheduling tools	Communicate delay to downstream teams; adjust production and QA release schedules; update batch records

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Production Purification Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Filtration failure or membrane rupture	Product loss, microbial contamination, batch rejection, deviation from hold-time limits	Medium	High	High	Production, QA, QC, Engineering	Use qualified filters, perform integrity testing (pre- and post-use); validate hold times and flow rates	Halt process, isolate product, notify QA; conduct RCA; retest if recovery possible; document deviation
Column packing irregularities in chromatography	Poor resolution, yield loss, impurity carryover, batch rejection	Medium	Medium	Medium	Production, QA, QC	Follow validated packing SOPs; perform packing qualification runs; monitor pressure and flow profiles	Stop purification, re-pack column; log deviation; QA to assess impact on purity and batch disposition
Contamination during buffer preparation or transfer	Microbial or particulate contamination, batch	Medium	High	High	Production, QA, Microbiology	Prepare buffers in controlled areas; use validated SOPs; closed-	Discard contaminated buffer, prepare fresh batch, notify QA and

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
	rejection, regulatory deviation					loop transfer systems; EMI of buffer tanks	production; initiate deviation investigation
Buffer mislabelling or incorrect connection during purification	Cross-contamination, process deviation, product quality impact	Low	High	Medium	Production, QA	Use color-coded lines and barcode labels; perform line clearance verification; double-check connections	Stop process; QA to assess cross-contamination risk; determine salvageability; document deviation
Equipment failure (chromatography skid, UF/DF system)	Process interruption, yield loss, extended downtime, batch rejection	Medium	Medium	Medium	Engineering, Production, QA	Maintain AMC; stock critical spares; perform periodic preventive maintenance; validate backup systems	Transfer process to backup system if available; escalate to Engineering and QA; assess impact and document

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Concentration step failure (diafiltration error, TMP imbalance)	Product degradation, altered concentration profile, dosing inaccuracy	Medium	Medium	Medium	Production, QA, QC	Real-time monitoring of pressure, flux, conductivity, validate flow paths and hold times	Assess concentration profile; QA to evaluate degradation potential; disposition decision; initiate CAPA
Temperature excursion during holding steps	Stability loss, potency degradation, regulatory non-compliance	Low	High	Medium	Production, QA, Engineering	Install temperature alarms; use validated jacketed vessels; integrate with BMS; monitor excursion logs	Transfer product to alternate controlled storage; evaluate impact on stability; QA to assess disposition
Cleaning validation failure or CIP/SIP inadequacy	Residual contamination, cross-batch risk, regulatory deviation	Low	High	Medium	QA, Engineering, Production	Perform routine CV checks; trend analysis; calibrate conductivity	Re-clean and revalidate; investigate failure source; QA to determine

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
						and TOC sensors; validate cleaning cycles	batch disposition; document deviation
Yield deviation impacting batch size or dosing accuracy	Inconsistent dosing, batch rejection, regulatory deviation	Medium	Medium	Medium	Production, QA, QC	Monitor step recovery, perform trending and process capability studies, reconcile material balance	Evaluate need for reprocessing or pooling, QA approval mandatory, document deviation and corrective action

Production Fill-Finish Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Incomplete filling, volume	Batch hold or rejection, regulatory deviation, sterility	Medium	High	High	Production, QA, QC, Regulatory Affairs	Maintain validated secondary fill line; perform preventive maintenance and calibration	Shift to validated manual filling SOP; inform QA; document deviation;

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variation, or aseptic breach	failure, dosing inaccuracy					of filling equipment; operator training on manual mode and aseptic technique	evaluate in-process control data for batch disposition
Camera or conveyor fault during 100% inspection	Defective vials escaping detection; product recall risk; regulatory non-compliance	Medium	High	High	QA, Production, Engineering	Periodic calibration of camera and lighting systems; preventive checks on reject ejection mechanism; maintain manual inspection SOP	Initiate manual visual inspection by qualified staff; record transfer in deviation log; resume automated inspection post QA-cleared rectification
Dispatch delay or mislabelling impacting traceability	Regulatory audit findings; product mix-up; recall risk; compliance deviation	Medium	Medium	Medium	QA, Warehouse, Regulatory Affairs	Maintain backup label printer; SOP for visual verification and reconciliation; regular	Switch to manual labelling with QA visual verification; record event in batch record; resume automated

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
						cleaning of printer head and sensors	labelling post repair with label verification
Filter integrity failure or absence of validated backup	Batch contamination, sterility breach, product rejection, regulatory deviation	Low	High	Medium	QA, Production, Engineering	Maintain validated sterile filters in backup inventory, on-site integrity test kit (pre-post-use), SOP training for filter integrity testing	Replace failed filter with validated backup; conduct integrity test immediately; QA to assess impact and disposition of material processed through affected filter

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Quality Assurance Batch Release Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Delay in product release due to reviewer unavailability or backlog	Supply disruption, missed market timelines, customer dissatisfaction, reputational impact	Medium	High	High	QA, Regulatory Affairs, Supply Chain	Cross-training of QA reviewers; defined batch review timelines; pre-dispatch QA checklist; reviewer workload balancing	Escalate to QA Manager for alternate reviewer; assignment; prioritize release based on product criticality (e.g., shortage, export commitment); document justification in batch record
Missed timelines leading to audit findings or regulatory non-compliance	Warning letters, inspection observations, reputational	Medium	High	High	QA, Regulatory Affairs, Senior Management	Electronic batch tracking system or Excel tracker; weekly review meetings with QA Head;	Notify Quality Head and Regulatory Affairs; expedite root cause analysis; implement interim controls; initiate

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
	damage, delayed approvals					escalation matrix for overdue reviews	CAPA and document deviation
Unavailability of SOPs, BMRs, or validation records during audits or operations	Audit non-compliance, delayed responses, regulatory penalties, operational confusion	Low	High	Medium	QA, Documentation, Cell, Regulatory Affairs	Controlled document repository (physical/digital); SOP for issuance and archival; periodic backup of digital records; access control logs	Retrieve from backup/archive, notify concerned user, re-issue controlled copies with justification and QA sign-off, log event in deviation register
Implementation of unapproved changes affecting compliance	Regulatory deviation, product quality impact, audit findings, recall risk	Low	High	Medium	QA, Production, Engineering, Regulatory Affairs	QA-driven change control process; impact assessment and approval matrix; monthly review	Freeze ongoing change if risk identified; QA to assess and escalate unauthorised change as

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
						of change log; training on change SOP	deviation; initiate impact assessment and CAPA
Regulatory non-compliance due to untrained personnel	Invalidated operations, audit observations, product quality risk, compliance breach	Medium	Medium	Medium	QA, HR, Production, Warehouse	Training matrix linked to job function; monthly review of overdue trainings; qualified trainers and assessment system; training SOP compliance	Restrict untrained personnel from GMP activity; conduct immediate refresher training; log justification and corrective action; update training records
Incomplete records or unprepared responses during internal/external audits	Audit delays, negative inspection outcomes, CAPA burden, reputational impact	Medium	Medium	Medium	QA, Documentation, All Departments	QA audit readiness calendar; periodic mock audits; CAPA closure tracking; QA audit coordinator nominated	Activate audit response team; provide corrective documentation and justifications; initiate

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
							CAPA for identified gaps; conduct post-audit review
Delayed response or regulatory action due to poor traceability or investigation	Product recall, regulatory action, market withdrawal, reputational damage	Low	High	Medium	QA, Regulatory Affairs, Supply Chain, Pharmacovigilance	Market complaint SOP; investigation flowchart; QA-led recall procedure; complaint trend monitoring and analysis	Activate recall committee; notify regulatory body if applicable; document all steps; execute recall as per SOP; conduct post-recall review and CAPA

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Quality Assurance QMS Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Missed deviation timelines or inadequate RCA	Repeat deviations, audit findings, regulatory observations, loss of credibility	Medium	High	High	QA, Production, Regulatory Affairs	Defined deviation timelines in SOP; deviation tracker with escalation matrix; reviewer/approver backups; RCA training	Escalate overdue deviations to QA Head; assign alternate reviewer; initiate risk-based interim controls; document justification and CAPA
Ineffective CAPA implementation	Recurrence of deviations, regulatory penalties, inspection failures	Medium	High	High	QA, All Operational Units, Regulatory Affairs	Effectiveness check matrix; periodic CAPA trend analysis; QA review and approval of all CAPAs; CAPA closure SOP	Re-open CAPA if failure observed; implement short-term corrective actions; update risk register; conduct fresh RCA; escalate to QA Head

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Unauthorized or poorly assessed changes	Product quality impact, regulatory deviation, audit findings, recall risk	Low	High	Medium	QA, Production, Engineering, Regulatory Affairs	Impact assessment checklist, cross-functional QA-initiated review, change control log with version tracking, change SOP training	Suspend change implementation, raise deviation for unauthorized change, QA to lead retrospective impact assessment, initiate CAPA.
Incomplete records or untrained staff during audits	Critical audit observations, inspection delays, compliance breach	Medium	Medium	Medium	QA, HR, Documentation Cell, All Departments	Monthly audit readiness checklist, QA-led internal audits, CAPA tracking, document retrieval SOP	Activate audit response team; retrieve missing records; QA to submit CAPA plan to regulatory authorities (if applicable); conduct post-audit review
Loss of controlled documents or use	Non-compliance, operational errors	Low	High	Medium	QA, Documentation	Master list of controlled documents, archival and	Re-issue documents with version control; deactivate

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
of obsolete versions	audit findings; regulatory penalties				Cell, All Departments	issuance SOP; digital access with audit trail; periodic document review	obsolete copies; log incident as deviation if released; notify impacted departments
Untrained or undertrained staff in GMP operations	GMP errors; batch deviations; product quality risk; audit observations	Medium	Medium	Medium	QA, HR, Production, Warehouse	Defined training matrix per department; assessment and retraining schedules; QA training SOP; training effectiveness review	Restrict access to GMP operations for untrained personnel; conduct immediate training; document justification and corrective action; update training records

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Quality Control Analytical Testing Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Testing delays due to instrument failure or overload	Inability to release raw materials or finished product on time; supply chain disruption; delayed production	Medium	High	High	QC, QA, Production, Supply Chain	Preventive Maintenance (PM) schedule; backup validated instruments; AMC with OEM/vendor; instrument usage log	Shift testing to backup instrument; inform QA and log deviation; reschedule testing priorities; document impact on release timelines
Invalid test results due to calibration or system suitability failure	Regulatory non-compliance; batch rejection; audit observations; data integrity concerns	Medium	High	High	QC, QA, Regulatory Affairs	Calibration schedule with QC review; daily system suitability checks; use of calibrated reference standards; analyst qualification	Quarantine affected test data; recalibrate and retest impacted samples; conduct root cause analysis; document deviation and CAPA

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Test failure, OOS results, or delay due to reagent/material issues	Delayed batch release; OOS investigations; vendor-related compliance risk	Medium	Medium	Medium	QC, QA, Procurement, Production	Inventory control with minimum stock alerts; COA and expiry verification; alternate vendor qualification; reagent tracking SOP	Inform QA and Production; arrange alternate batch/vendor; document impact and retest if required; initiate deviation if delay affects release
Data integrity issues (e.g., transcription errors, unauthorized edits)	Batch rejection; regulatory penalties; audit findings; reputational damage	Low	High	Medium	QC, QA, IT, Regulatory Affairs	Analyst training and qualification; periodic assessments; double verification of entries; use of LIMS and audit trails	Isolate affected data; initiate deviation; re-train analyst if needed; QA to review re-testing justification; update data integrity log

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Data loss due to system failure or power outage	Halted testing activity, delayed release, loss of traceability	Low	High	Medium	QC, QA, IT	UPS for critical systems; regular data backups; manual data entry templates as contingency; IT SOP for recovery	Use manual logbooks/forms temporarily; resume LIMS entry post-restoration; IT to validate data integrity; QA to assess impact on batch disposition
Compromised product samples or study data due to chamber failure	Invalid stability data; regulatory reporting; batch rejection; audit concern	Low	High	Medium	QC Stability; QA, Regulatory Affairs	Environmental monitoring with alarms; backup qualified chambers; periodic mapping and validation; chamber qualification SOP	Shift samples to alternate chamber; QA to assess data impact; notify regulatory bodies if required; document deviation and initiate CAPA

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
False positive results due to contamination or EM failure	Test repetition; audit concern; delayed release; potential batch rejection	Medium	Medium	Medium	QC Microbiology, QA, Engineering	Regular EM; cleaning validation; gowning compliance; unidirectional flow and pressure control; EM trending	Suspend affected area; resample and retest; conduct full investigation; trend EM data; initiate CAPA if contamination confirmed
Increased error rate due to analyst competency gaps	Audit findings; invalid results; batch disposition delays; regulatory non-compliance	Medium	Medium	Medium	QC, QA, HR	Analyst training matrix; method-specific qualification; review of test records before approval; periodic competency assessments	Suspend non-compliant testing; train/qualify personnel before resumption; QA to review impact on prior data; document retraining and justification

Microbiology – EM & Microbiological Testing Risk Management Plan

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Missed EM data points or undetected contamination	Incomplete trending, compromised area qualification, undetected contamination risk, audit findings	Medium	High	High	QC, Microbiology, QA, Production	Defined EM schedule; trained EM team; use of pre-labelled EM plates; calibrated equipment; EM SOP compliance	Re-perform missed monitoring if within allowable window; log deviation; assess impact on batch and area status; increase EM frequency temporarily
Risk to sterile product quality due to EM excursion	Batch rejection, regulatory non-compliance, product recall, reputational damage	Medium	High	High	QC, Microbiology, QA, Regulatory Affairs, Production	Routine EM trending and alert limits; validated cleaning procedures; qualified disinfectants; cleanroom behaviour training	Hold affected product; increase EM frequency; investigate source; perform sanitization and requalification

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
							document deviation and CAPA
Invalid results due to temperature/humidity excursions or data loss	Inaccurate EM data, compromised incubation, invalid test results, audit concerns	Low	High	Medium	QC, Microbiology, QA, Engineering, IT	Monitoring with alarms, equipment qualification, backup incubators with validation, environmental SOPs	Quarantine affected samples, assess impact on result validity, initiate deviation, resample where appropriate, document incident and corrective action
False results due to unsterile or expired media	Test repetition, invalid data, audit findings, delayed batch release	Medium	Medium	Medium	QC, Microbiology, QA, Stores	Media GPT, COA and expiry verification, segregated storage,	Discard affected media, resample and retest with new media, record deviation, inform QA,

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
						controlled issuance; media inventory SOP	trend media-related deviations
False positives or missed true contamination	Unnecessary investigations; missed contamination; misleading EM trends	Medium	Medium	Medium	QC Microbiology, QA	Validated sampling procedures; use of disinfectant neutralizers; aseptic handling by trained staff; EM SOP adherence	Confirm source through re-sampling and parallel controls; trend historical data; perform EM impact assessment; initiate CAPA if needed
Misleading investigation outcomes or incorrect RCA	Inaccurate CAPA; recurrence of contamination; audit observations	Low	High	Medium	QC Microbiology, QA, Regulatory Affairs	Validated identification systems (e.g., VITEK, MALDI-TOF); periodic analyst proficiency	Re-identify isolate using validated method; update investigation report;

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
						testing; SOP for microbial identification	document correction; revise CAPA if needed
Batch hold or QA approval delay due to testing backlog	Delayed product release, supply chain disruption, audit trail gaps	Medium	Medium	Medium	QC, Microbiology, QA, Production, HR	Workforce planning; shift-based allocation; advance media planning; EM workload balancing	Notify QA; reschedule testing priorities; deploy trained backups; document impact if batch is delayed; initiate deviation if timelines breached
GMP non-compliance during audits or inspections	Regulatory observations, warning letters, reputational impact	Medium	High	High	QC, Microbiology, QA, Regulatory Affairs	Use of EM templates/logbooks with cross-checks; periodic QA review of EM records;	Audit trail review (electronic); manual investigation and justification (paper-based); escalate to QA

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Scenario	Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
						transition to electronic systems with audit trails	for CAPA; conduct post-audit review

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Regulatory Affairs – Dossier Preparation & Submission Risk Management Plan

Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Missed submission deadlines	Delayed product approval or launch; market entry loss; reputational impact	Medium	High	High	Regulatory Affairs, QA, QC, Production, Project Management	Maintain regulatory filing calendar with buffer timelines; cross-functional tracking of Module 1-5 inputs; periodic RA status meetings	Escalate delay to RA Head; prioritize high-risk submissions; communicate revised timelines to stakeholders; document deviation
Deficiency letters or multiple queries from authorities	Risk of rejection; extended review cycle; additional cost and resource burden	Medium	High	High	Regulatory Affairs, QA, QC, Production	Checklist-based review system; cross-verification of modules; pre-submission technical review; historical query analysis	Conduct root cause analysis; revise affected modules; provide clarification to authority; initiate CAPA and update SOPs

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Regulatory non-acceptance or validation delay	Submission rejection; reformatting burden; delayed review initiation	Low	High	Medium	Regulatory Affairs, IT	Use validated eCTD software; country-specific templates; maintain updated regulatory guidelines and checklists	Reformat and revalidate modules; resubmit per authority requirements; initiate corrective training for RA team; log deviation
Submission of outdated or conflicting documents	Regulatory queries; credibility loss; potential rejection	Medium	Medium	Medium	Regulatory Affairs, QA, Documentation Cell	Document control via version tracking; secure shared platforms (RA database, EDMS); SOP for document review	Identify incorrect version; submit corrigendum; document deviation and corrective action; update document control SOP
Missed submission window or rejected upload	Delay in regulatory acceptance; loss of priority review	Low	High	Medium	Regulatory Affairs, IT	Backup eCTD tools; trained RA personnel; IT support for portals; dry runs before submission	Use manual submission route (USB/CD if applicable); inform regulatory authority

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Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
	slot; reputational impact						proactively; document incident and resolution
Delay in receiving cross-functional data	Incomplete dossier; missed timelines; fragmented submission	Medium	Medium	Medium	Regulatory Affairs, QA, QC, Clinical, Packaging, Project Management	Shared regulatory tracker; inter-departmental meetings; pre-submission readiness checklist; owner mapping	Escalate to site head or project manager; fast-track missing inputs via dedicated support; document impact and mitigation
Withdrawal of application or major delay in review cycle	Loss of regulatory momentum; reputational damage; resource wastage	Low	High	Medium	Regulatory Affairs, QA, Senior Management	Query response SOP with owner mapping; response timelines; archive past query templates for learning	Allocate query response owner immediately; prepare draft within 24-48 hours; escalate data gaps to senior RA lead; document response strategy

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R&D – Process Development & Tech Transfer Risk Management Matrix

Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Poor scalability, batch failures, or process deviations at manufacturing site	High	High	Critical	R&D, Manufacturing, QA	- DoE and QbD-based process design - Lab-scale and pilot trials - Cross-functional design reviews	- Revisit CPPs and CQA alignment - Conduct RCA and CAPA - Revise development report and TT package
Gaps in regulatory dossiers and site readiness, delays in validation and filing	Medium	High	High	R&D, RA, QA, Manufacturing	- Maintain structured technical reports - Version-controlled development history - Standardized TT templates	- Reconstruct missing documentation - Engage SMEs for technical gap closure - Update validation and filing timelines

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Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Process non-reproducibility, analytical variability, regulatory non-compliance	Medium	High	High	R&D, QC, QA, RA	- Structured IT protocol and checklist - Training at receiving site - Trial batches with concurrent QA review	- Halt scale-up - Initiate risk-based deviation - Retraining and on-site R&D support
Critical parameter shift (e.g., mixing, heat transfer), product quality impact	Medium	Medium	Moderate	R&D, Manufacturing, QA	- Equipment equivalence mapping - Simulation and gap assessment studies	- Modify process parameters (RPM, feed rate) - Conduct bridging study or validation batch
Regulatory rejection, repeat studies, loss of trust in data	Low	High	High	R&D, RA, QA	- Audit trails and ELN/LIMS systems - Secure raw data	- Recover archived data - Reconstruct study with justification memo

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Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
					backups - R&D-GMP bridge training	- Notify RA for filing updates
Postponed exhibit batches or commercial launch	Medium	High	High	R&D, RA, Manufacturing, Commercial	- Project tracker with milestone alignment - Site-readiness checklist for validation	- Escalate to TI R&D head - Re-align timelines and regulatory deliverables
QC/QA unable to reproduce R&D test methods	Medium	Medium	Moderate	R&D, QC, QA	- Pre-approved method transfer protocol - Co-validation or comparative testing	- Provide R&D technical support - Amend SOPs - Retrain receiving lab and document changes
Deficiency letter from regulatory agencies or post-approval variation	Low	High	High	RA, R&D, QA	- Early RA alignment during development	- Amend Module 3 sections

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Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
					- Filing strategy discussions	- File variation if required - Document in TT closure report

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IT Risk Management Matrix – GxP Systems & Compliance

Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Loss of controlled procedures (SOPs, protocols), delayed QA reviews or document issuance	Medium	High	High	QA, IT, Document Control	- Role-based access control - Enabled audit trails and version control - Periodic system health checks and SOP backups	- Activate fallback SOP repository (hardcopy) - Initiate manual issuance log - Notify QA and track restoration status
Training delays, inability to assign or record compliance training, potential for audit findings	Medium	Medium	Moderate	HR, QA, IT, Training	- Maintain validated LMS with user matrix - Periodic data backups - SOPs for training assignment and tracking	- Track overdue trainings manually - Maintain signed offline records - Escalate to HR and QA with compliance justification

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Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Permanent loss of GMP records, audit trails, or user activity logs affecting data integrity	Low	High	High	QA, IT, RA	- Scheduled daily/weekly backups - Backup verification tests - 3-tier architecture (Local-NAS-Cloud)	- Initiate disaster recovery protocol - Restore from latest verified backup - Document recovery status and data gaps
Inaccessibility of GxP systems (DMS, LMS, SCADA) affecting batch review, training, and reporting	Medium	High	High	Manufacturing, QA, IT, Training	- Redundant servers and mirrored storage - UPS and DG backup power - Preventive server maintenance	- Notify system users - Switch to manual SOP-based process - Resume digital entry post-restoration
Potential non-compliance with Part 11/Annex 11, audit observations	Medium	High	High	QA, IT, RA	- Maintain validation calendar	- Restrict new updates until re-validation

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Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
					- Periodic user access reviews - Regualification plans for updates	- Document deviation and justification - QA-approved memo for audit trail
Data compromise, system lockdown, unauthorized access to confidential or regulated data	Low	High	High	IT, QA, Data Integrity, RA	- Antivirus and firewall protection - Restricted admin rights - Incident response SOP and monitoring	- Isolate affected systems - Activate IT Security SOP - Notify Data Integrity Officer and QA - Conduct forensic analysis
Unauthorized data modification or training bypass	Medium	Medium	Moderate	QA, IT, HR, Training	- Periodic access reviews with department heads - Auto-deactivation of	- Audit access logs - Deactivate non-compliant accounts

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Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
					idle users - Role-based access assignment	- Raise deviation and document unauthorized activity

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Warehouse & Cold Chain Risk Management Matrix

Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Inability to track stock movement, delayed dispatches, stock reconciliation issues	Medium	Medium	Moderate	Warehouse, IT, QA, Planning	- Regular ERP system backups - SOP for manual inventory recording - Staff trained on fallback process	- Shift to manual logbooks - Track critical transactions manually - Reconcile data post-restoration and document gaps
Risk of batch failure or recall due to wrong API/excipient dispensing	Low	High	High	Warehouse, QA, Production	- FIFO/FEFO practices - Barcode/RFID tracking - Monthly stock audits and reconciliation	- Quarantine affected batch - Conduct internal investigation - Reissue stock under QA supervision with deviation record

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Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Product temperature excursion, vaccine/biologic degradation	Medium	High	High	Warehouse, QA, RA	- Continuous temperature monitoring with alerts - Backup cold storage units - Calibrated sensors and alarm testing	- Transfer to alternate qualified unit - Notify QA immediately - Assess excursion impact and disposition decision
Temperature deviation in cold rooms and freezers	Medium	Medium	Moderate	Warehouse, Engineering, QA	- DG backup for refrigeration - UPS for monitoring devices - Daily cold storage status checks	- Activate alternate power supply - Document excursion - Use thermal mapping data to assess product impact

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Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Cold chain breach during distribution to customers sites	Low	High	High	Logistics, QA, RA, Warehouse	- Validated thermal packaging - GPS/loggers in transport boxes - Qualified logistics partners	- Inform recipient site - Quarantine shipment on arrival - Review logger data and investigate transporter
Cross-contamination, chemical reactivity, regulatory non-compliance	Low	High	High	Warehouse, EHS, QA	- Color-coded zoning - MSDS availability - SOP-based material segregation and handling	- Isolate and relabel material - Perform area decontamination - Raise deviation and conduct risk assessment
Product may be used without stability assurance	Medium	High	High	QA, RA, Warehouse	- Daily shift-wise temperature review - Alarm testing	- Notify QA and RA - Block impacted stock

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Potential Impact	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
					- Periodic sensor calibration	- Perform stability impact evaluation and raise deviation
Batch delays, missed dispatch schedules	High	Medium	High	Planning, Procurement, Warehouse	- Minimum stock level triggers - Weekly MRP review - Alternate vendor qualification	- Escalate to Planning and Procurement - Initiate emergency procurement - Seek substitution approval if applicable

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Workforce Availability & Safety Risk Management Matrix

Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Staff Shortage in Critical Functions	Production halt, QA delays, compromised support services, missed RTO targets	Medium	High	High	Production, QA, HR, Admin	Workforce redundancy, shift rotation, transport tie-ups	Activate staffing matrix, notify HODs, arrange alternate transport/lodging
Lack of Qualified Operators	Batch deviation, regulatory non-compliance, quality risk	Medium	High	High	Production, QA, HR	Validated training matrix, cross-functional backups, retraining	Deploy backups, escalate to QA, document deviation
Workplace Accident or Safety Breach	Legal exposure, morale dip, operational stoppage, reputational risk	Low	High	Medium	All departments, EHS, HR	PPE enforcement, safety drills, ERT readiness, SOP adherence	First aid, isolate area, investigate, notify EHS, HR, report externally if needed

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Harassment or Grievance Incident	Legal liability, attrition, team disruption, audit non-conformance	Low	Medium	Medium	HR, Admin, All departments	POSH training, grievance committee, anonymous complaint system	Formal inquiry, counselling, disciplinary action per HR policy
Site Access Disruption	Missed shifts, security breach, production delays	Medium	Medium	Medium	Security, Admin, Production, HR	Alternate access routes, transport tie-ups, emergency contact list	Inform security/ERT, activate alternate gate, notify staff via emergency channels
Delayed Evacuation or Rescue	Safety risk, regulatory breach, delayed emergency response	Low	High	Medium	EHS, Security, HR	Updated emergency contact list, monthly verification	Use physical muster rolls, coordinate headcount with leads/security

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Training Deficiencies	Audit findings, restricted operations, product quality risk	Medium	Medium	Medium	QA, HR, Production	LMS tracking, refresher sessions, training audits	Conduct training, restrict untrained staff from GMP zones, document deviation
Unauthorized Access to Sensitive Areas	IP theft, contamination risk, sabotage, audit non-compliance	Low	High	Medium	Security, R&D, Production, Admin	24x7 manned posts, access control logs	Alert Security Supervisor, restrict access, initiate internal review
Theft or Sabotage of Materials/Documents	Financial loss, production disruption, legal exposure	Low	High	Medium	Stores, QA, Admin, Legal	ID-based access control, visitor logs	Conduct internal investigation, escalate to Legal if needed
Loss of Confidential Information	IP leakage, reputational damage, regulatory scrutiny	Low	High	Medium	QA, IT, Legal, R&D	Biometric RFID access zones, data protection SOPs	Notify QA/IT/Legal, initiate incident handling protocol

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Vandalism During Off-Hours	Infrastructure damage, operational delay, forensic liability	Low	Medium	Medium	Admin, Security, Legal	CCTV surveillance, perimeter monitoring	Seal area, initiate forensic investigation, report to leadership
Physical Threat to Employees	Safety risk, legal action, operational disruption	Low	High	Medium	HR, Security, Admin	Background checks, security clearance for contractors	Evaluate affected staff, involve local police, document incident
Delayed Emergency Response	Escalation of fire/riot/intrusion, safety breach	Medium	High	High	EHS, Security, Admin	Emergency drills, ERT coordination	Trigger ERP, notify authorities, log response timeline
Poor Record-Keeping of Movements	Investigation delays, audit findings, security gaps	Medium	Medium	Medium	Admin, Security, HR	Vehicle registers, visitor badges, gate pass system	Cross-verify logs, escalate anomalies, update SOPs

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Reputational Damage from Security Lapses	Loss of stakeholder trust, audit penalties, media exposure	Low	High	Medium	Admin, Security, Corporate Affairs	Security SOPs, periodic audits	Submit incident report, implement CAPA, brief leadership

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Fire Risk Management Matrix – Safety & EHS

Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Fire in Solvent Handling or Electrical Zones	Employee injury/fatality, toxic exposure, legal liability, shutdown of hazardous zones	Medium	High	High	Production, EHS, HR, Engineering	Fire extinguishers/hydrants installed; monthly checks; solvent handling SOPs	Shut off energy sources (electricity, gas, steam); ERT to guide evacuation and operate extinguishers if trained
Fire Damage to Production/QC/Utility Equipment	Equipment loss; extended downtime; batch delays; CAPEX impact	Medium	High	High	Production, QC, Engineering, Maintenance	Smoke detectors and alarms in all key zones; electrical load audits	Engage main shut-off switches; notify fire department; assemble in safe zones; initiate damage assessment

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Batch Failure or Contamination from Fire/Smoke Suppression Agents	Product loss, deviation reports, regulatory non-compliance	Low	Medium	Medium	QA, Production, EHS	Flameproof fixtures, ATEX-rated equipment, electrical zoning	Restrict re-entry until cleared; conduct headcount; QA to assess batch integrity
Regulatory Non-Compliance or Site Shutdown	Penalties, audit findings, reputational damage, license suspension	Low	High	Medium	Corporate EHS, QA, Admin	Emergency exit maps, escape route signage, fire SOPs	Document cause and extent; report to corporate EHS; initiate CAPA
Contamination of Sterile Areas by Suppression Agents	Sterility breach, batch rejection, deviation, requalification	Low	High	Medium	QA, Production, Engineering	Heat-risk sensors, auto shut-off systems, zoning of suppression types	Resume operations only after clearance from QA, EHS, and Engineering;

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
							document requalification
Business Continuity Disruption Due to Fire Incident	Delivery delays, customer impact, reputational loss, insurance claims	Medium	High	High	Supply Chain, QA, Corporate Affairs	Quarterly fire drills, puncture-proof cabinets, overload protection	Align investigation with incident SOP; escalate to leadership; initiate post-incident mitigation and CAPA
Fire Spread Due to Inadequate Containment	Multi-zone damage, cross-contamination, extended downtime	Low	High	Medium	EHS, Engineering, Admin	Fire-rated doors, containment barriers, zoning protocols	Activate containment SOP; isolate affected zones; coordinate with fire officials

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Delayed Emergency Response or Evacuation	Increased injury risk, regulatory breach, panic escalation	Medium	High	High	EHS, Security, HR	Emergency response drills, ERT training, evacuation maps	Trigger ERP; notify authorities; conduct headcount; log response time
Failure to Maintain Fire Safety Equipment	Non-functional extinguishers, audit findings, increased risk	Medium	Medium	Medium	EHS, Maintenance, Admin	Monthly inspection logs, third-party audits, calibration records	Replace/repair faulty equipment; document non-conformance; retrain staff
Improper Storage of Flammable Chemicals	Fire ignition, explosion risk, regulatory violation	Medium	High	High	Stores, Production, EHS	Puncture-proof cabinets, segregation protocols, MSDS availability	Isolate affected area; notify EHS; initiate root cause analysis and CAPA

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Flood Risk Management Matrix – Safety, EHS & Operations

Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Water Ingress into Cleanrooms, Production, Warehouse, Electrical Panels	Equipment damage, sterility breach, production halt, electrical hazards	Medium	High	High	Production, QA, Engineering, EHS	Install sump pumps, auto-drainage systems; monthly testing	Activate Emergency Response SOP, isolate power, ERT to guide evacuation
Spoilage or Loss of RM/FG Due to Low Storage Elevation	Financial loss, batch delays, deviation reports, insurance claims	Medium	High	High	Stores, QA, Production, Supply Chain	Elevate inventories and panels above flood level; waterproof storage zones	Secure entry points with sandbags/plastic barriers; initiate salvage protocol
Disruption of HVAC, Compressed Air, Utilities	Environmental control failure, sterility risk, equipment shutdown	Medium	High	High	Engineering, Production, EHS	Seal utility wall inlets (cable trays, pipelines); waterproofing audits	Evacuate non-essential staff, confirm headcounts; isolate affected systems

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Cold Storage Temperature Excursions Due to Power/Insulation Failure	Product degradation, deviation, regulatory non-compliance	Medium	High	High	QA, Engineering, Stores	Emergency lighting, illuminated escape signs, insulation audits	Coordinate with Engineering for backup power, initiate isolation SOP
Contamination in Sterile/Microbiology Areas	Batch rejection, requalification, audit findings	Low	High	Medium	QA, Microbiology, Production	Weather monitoring apps, municipal alerts, early warning protocols	Isolate and label exposed inventory; photograph for traceability; QA to assess
Halt in Batch Release/QA Approvals Due to Data Inaccessibility	Regulatory delays, audit non-conformance, operational bottlenecks	Medium	Medium	Medium	QA, IT, Regulatory Affairs	Cloud backup of critical data; external server redundancy	Manual transfer of data/documents; escalate to IT for recovery support

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Logistics Disruption Due to Blocked Roads or Gate Flooding	Delivery delays, customer impact, supply chain breakdown	High	Medium	High	Supply Chain, Admin, Security	Pre-map alternate routes; share contingency plans with logistics teams	Coordinate with local authorities for access/rescue; reroute shipments
Flood-Induced Electrical Short Circuits or Fire Hazards	Safety risk, equipment loss, legal exposure	Medium	High	High	Engineering, EHS, Admin	Waterproof electrical panels; install surge protection; conduct audits	Shut down affected circuits; notify ERT; initiate root cause analysis
Delayed Response Due to Lack of SOP Familiarity or Drill Gaps	Escalated damage, safety risk, audit findings	Medium	Medium	Medium	EHS, HR, All Departments	Quarterly flood drills; SOP refresher sessions; ERT role clarity	Trigger ERP; conduct headcount; log response time and gaps for CAPA

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Reputational Damage Due to Poor Flood Preparedness	Stakeholder trust erosion, media exposure, regulatory scrutiny	Low	High	Medium	Corporate Affairs, QA, EHS	Public-facing SOPs, leadership briefings, CAPA visibility	Submit incident report, initiate mitigation plan, communicate transparently

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Earthquake Risk Management Matrix – Safety, EHS & Operations

Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Structural Damage to Production, QC, Utility Blocks, Warehouses	Operational downtime, equipment loss, batch delays, regulatory non-compliance	Medium	High	High	Engineering, Production, QA, Admin	Compliance with seismic construction codes; structural audits	Activate Emergency Response SOP, evacuate via marked exits; initiate damage assessment
Toppling of Critical Equipment (e.g., Gas Cylinders, Autoclaves)	Explosion risk, fire hazard, injury, contamination	Medium	High	High	Engineering, EHS, Production	Periodic structural inspections, secure anchoring of heavy equipment	ERT to assess damage; isolate hazardous zones; coordinate emergency response
Loss of Cleanroom Integrity Due to AHU or Structural Damage	Sterility breach, batch rejection, requalification delays	Low	High	Medium	QA, Microbiology, Engineering	Anchor movable equipment; reinforce ceiling/floor structures	Shut down utilities; isolate cleanrooms; QA to assess requalification needs

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Personal Injury from Falling Debris or Equipment Collapse	Fatality risk, legal liability, morale impact	Medium	High	High	HR, EHS, All Departments	Seismic-resistant racks, safe storage protocols	Conduct roll call, provide first aid, escalate to medical and HR teams
Product Sample Damage from Unstable Storage	Batch loss, deviation reports, audit findings	Medium	Medium	Medium	QA, Stores, R&D	Anchor shelves, secure sample storage, SOP adherence	Inspect storage zones, document damage, QA to assess salvageability
Manufacturing Downtime Pending Structural Clearance	Delivery delays, customer impact, financial loss	Medium	High	High	Production, QA, Engineering	Emergency exit signage, semi-annual evacuation drills	Re-enter only after clearance from civil engineering team; document downtime
Delayed Emergency Support Due to	Rescue delays, extended downtime, reputational risk	Medium	High	High	Admin, Security, Supply Chain	Emergency shut-off valves, alternate	Maintain communication with authorities; activate alternate logistics routes

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Roadblocks or Infrastructure Damage						recovery site planning	
Loss of Utility Services (Electricity, Gas, Steam)	Equipment shutdown, sterility risk, operational halt	Medium	High	High	Engineering, Production, QA	Install auto shut-off valves; utility isolation SOPs	Shut down live utilities; coordinate with Engineering for restoration
Inaccessibility of Critical Documentation or Data	Regulatory delays, audit non-conformance	Low	Medium	Medium	QA, IT, Regulatory Affairs	Cloud backups; offsite data redundancy	Manual data recovery; escalate to IT; document deviation
Reputational Damage Due to Poor Earthquake Preparedness	Stakeholder trust erosion, media exposure, compliance risk	Low	High	Medium	Corporate Affairs, QA, EHS	Public-facing SOPs; leadership briefings; CAPA visibility	Submit incident report; initiate mitigation plan; communicate transparently

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Pandemic Risk Management Matrix – HR, EHS, QA & Operations

Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Infected or Quarantined Staff	Operational disruption, missed RTOs, batch delays, regulatory deviation	High	High	High	HR, Production, QA, Admin	Workforce segregation, remote work for non-GMP roles, health declarations	Activate critical role matrix, deploy trained backups, approve movement passes
Workplace Infection Spread	Site shutdown, legal exposure, reputational damage	Medium	High	High	EHS, HR, Admin, QA	Mandatory PPE, staggered shifts, sanitization, awareness training	Isolate suspected cases, sanitize zones, notify health authorities
Batch Manufacturing Halt Due to Input Shortage	Product delays, deviation reports, customer impact	Medium	High	High	Supply Chain, QA, Production	Alternate vendors, buffer stock, dual source qualification	Prioritize batches, inform Regulatory Affairs, activate alternate procurement

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Delayed Delivery to Hospitals/Markets	Product stockout, reputational risk, patient impact	Medium	High	High	Supply Chain, Commercial, Admin	Multiple logistics partners, emergency delivery routes	Inform customers, reroute via alternate partners, escalate to supply chain head
Postponed GMP Approvals or Certifications	Regulatory delays, audit findings, market access issues	Medium	Medium	Medium	QA, Regulatory Affairs, IT	Virtual submission readiness, eCTD systems, ongoing agency communication	Submit virtual documentation, request deferral or virtual audit
Inadequate Sanitization	Infection risk, audit non-compliance, morale impact	Medium	Medium	Medium	EHS, Admin, QA	Buffer stock of disinfectants, masks, thermal guns, multi-vendor sourcing	Prioritize GMP admin zones; initiate emergency purchase via admin team

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Stress-Related Absenteeism	Reduced productivity, morale dip, HR escalations	High	Medium	High	HR, Admin, All Departments	Counselling programs, rotational off-days, HR support lines	Provide psychological first aid, allow compassionate leave, engage counsellors
Batch Contamination Due to Health Breach	Product rejection, regulatory non-compliance, requalification	Low	High	Medium	QA, Production, Microbiology	Separate access/change rooms, health checks, gowning SOPs	Halt operations, sanitize per SOP, resume only after QA clearance
Delayed Emergency Response or Communication Breakdown	Escalation of infection, panic, regulatory breach	Medium	Medium	Medium	ERT, HR, Admin	Emergency communication protocols, ERT training, contact list updates	Trigger ERP, notify stakeholders, log response timeline for CAPA

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Risk Scenario	Potential Impacts	Likelihood	Impact Severity	Risk Rating	Affected Departments	Preventive Measures	Response Plan
Reputational Damage Due to Poor Pandemic Preparedness	Stakeholder trust erosion, media exposure, compliance risk	Low	High	Medium	Corporate Affairs, QA, HR	Public-facing SOPs, leadership briefings, CAPA visibility	Submit incident report, initiate mitigation plan, communicate transparently

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Impact Evaluation Matrix: Critical Business Functions & Scenarios

Each Critical Business Function (CBF) has been systematically assessed across four key impact dimensions: financial performance, operational continuity, regulatory compliance, and organizational reputation. This multi-dimensional evaluation ensures that the organization can prioritize its response and recovery efforts based on the severity and breadth of potential disruption.

The assigned impact levels—High, Medium, and Low—serve as qualitative indicators of risk exposure and business vulnerability.

Department / CBF	Scenario / Risk Event	Financial Impact	Business Disruption	Compliance Impact	Reputational Impact
Engineering – HVAC	Cleanroom pressure cascade failure	Medium	High	High	Medium
Engineering – Power	DG set failure during sterile operations	High	High	High	High
Engineering – Boiler	Steam supply interruption	High	High	High	Medium
Engineering – Briquettes	Shortage affecting sterilization and HVAC	Medium	High	High	Medium
Engineering – Cooling Tower	Microbial contamination (e.g., Legionella)	Medium	High	High	High
Engineering – Chiller	Temperature control failure in cleanrooms	High	High	High	High
Production – Upstream	Contamination in fermentation vessels	High	High	High	High

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Department / CBF	Scenario / Risk Event	Financial Impact	Business Disruption	Compliance Impact	Reputational Impact		
Production – Downstream	Filtration failure or membrane rupture	High	High	High	High		
Production – Fill-Finish	Aseptic breach or volume variation	High	High	High	High		
QA – Batch Release	Reviewer unavailability causing release delays	High	High	High	High		
QA – QMS	Missed deviation timelines or ineffective CAPA	Medium	Medium	High	Medium		
QC – Analytical Testing	Instrument failure delaying testing	High	High	High	Medium		
Microbiology – EM & Testing	EM excursion impacting sterile product	High	High	High	High		
Regulatory Affairs	Missed submission deadlines	High	Medium	High	High		
R&D – Tech Transfer	Poor scalability or process deviation	High	High	High	High		
IT – GxP Systems	Data loss or system failure affecting batch review	High	High	High	High		

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Department / CBF		Scenario / Risk Event		Financial Impact	Business Disruption	Compliance Impact	Reputational Impact
Warehouse & Logistics		Cold chain breach during distribution		High	High	High	High
HR / Workforce Availability		Staff shortage in critical functions		High	High	Medium	Medium
Safety & EHS – Fire		Fire in solvent/electrical zones		High	High	High	High
Safety & EHS – Flood		Water ingress into cleanrooms or electrical panels		High	High	High	High
Safety & EHS – Earthquake		Structural damage to production/QC blocks		High	High	High	High
Safety & EHS – Pandemic		Workplace infection spread		High	High	High	High

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Recovery Metrics Evaluation by Critical Business Function

Recovery Time Objective (RTO), Recovery Point Objective (RPO), and Maximum Tolerable Downtime (MTD) have been systematically defined and aligned for all critical business functions across XXX (Firm Name) Limited' operational landscape.

These recovery parameters are calibrated to ensure operational resilience, regulatory compliance, and continuity of essential services. Each function's RTO, RPO, and MTD has been benchmarked against ISO 22301, GMP standards, and internal risk tolerance thresholds, enabling targeted recovery strategies and escalation protocols.

Department / Function	Example Scenarios	RTO (Max Time to Recover)	RPO (Max Data Loss Tolerated)	MTD (Max Downtime Tolerated)
Production (Upstream)	Fermenter failure, media contamination, power outage	4-5 hours	1 hour	8-12 hours
Production (Downstream)	Filtration failure, chromatography deviation, buffer contamination	6-8 hours	2 hours	12-16 hours
Fill-Finish Operations	Aseptic breach, filling line failure, inspection system fault	2-4 hours	30 minutes	6-8 hours
Quality Assurance (QA)	Batch release delays, document unavailability, reviewer absence	8 hours	2 hours	24 hours

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Department / Function	Example Scenarios	RTO (Max Time to Recover)	RPO (Max Data Loss Tolerated)	MTD (Max Downtime Tolerated)	
Quality Control (QC)	Instrument failure, calibration issues, reagent stockouts	6–8 hours	1–2 hours	12–16 hours	
Microbiology / EM	Missed EM points, sterility test failure, media contamination	4–6 hours	1 hour	8–12 hours	
Regulatory Affairs	Submission delays, data integrity issues, query response gaps	24 hours	4 hours	48–72 hours	
Research & Development	Tech transfer delays, process reproducibility issues	24–48 hours	4–6 hours	72 hours	
Engineering & Utilities	HVAC failure, boiler outage, briquette shortage, chiller breakdown	2–4 hours	N/A (non-data function)	6–8 hours	
IT & Digital Systems	DMS/LMS failure, cybersecurity breach, data loss	1–2 hours	15–30 minutes	4–6 hours	
Warehouse & Logistics	Cold chain breach, inventory mismanagement, dispatch delays	4–6 hours	1 hour	12–16 hours	

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Department / Function	Example Scenarios	RTO (Max Time to Recover)	RPO (Max Data Loss Tolerated)	MTD (Max Downtime Tolerated)
Safety & EHS	Fire, chemical spill, evacuation delays	Immediate (<1 hour)	N/A	2-4 hours
HR & Admin	Workforce unavailability, access disruption, training gaps	6-8 hours	2 hours	12-24 hours

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BCP Activation Scenario Testing Framework

To ensure the effectiveness, reliability, and organizational readiness of XXX (Firm Name) Limited' Business Continuity Plan (BCP), a structured, risk-based testing program shall be implemented across all critical business functions. This program is designed to validate response capabilities, reinforce ownership, and demonstrate compliance with ISO 22301 and GMP expectations.

Objectives of BCP Testing

Each test scenario is designed to:

- ✔ Validate BCP activation protocols under simulated disruption conditions
- ✔ Ensure clarity of roles and responsibilities across departments and continuity leads
- ✔ Confirm resource availability and readiness, including personnel, infrastructure, and documentation
- ✔ Identify process and communication gaps that may hinder timely recovery
- ✔ Demonstrate regulatory compliance with ISO 22301, GMP, and internal QMS standards

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Types of BCP Tests

Test Type	Description	Primary Focus Areas
Tabletop Exercises	Structured walkthroughs of scenario-based disruptions with key personnel	Decision-making, escalation flow, procedural adequacy
Functional Drills	Targeted execution of departmental or system-level recovery processes	Technical response, SOP adherence, system resilience
Full-Scale Simulations	Site-wide mock drills simulating multi-departmental crisis scenarios (e.g., fire, pandemic)	Cross-functional coordination, external agency roles
Call Tree / Communication Drills	Periodic verification of emergency contact cascades and response times	Communication integrity, escalation timing

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Test Frequency and Coverage

Scenario Type	Minimum Test Frequency	Remarks
High-risk incidents (fire, flood, IT outage)	Annual	Conducted per site; includes utility failure simulations
Pandemic / workforce shortage	Every 3 years	Triggered by regulatory advisories or internal risk flags
Vendor disruption / supply chain failure	Annual	Coordinated with QA and strategic vendor partners
Cybersecurity breach / SCADA-PLC failure	Every 2 years	Includes incident simulation and digital recovery drills

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BCP Training & Awareness Framework

To ensure organizational readiness, cross-functional alignment, and regulatory compliance, XXX (Firm Name) Limited shall implement a structured Business Continuity Plan (BCP) training program. This program is designed to embed continuity principles into daily operations, reinforce scenario-specific preparedness, and cultivate a culture of resilience across all levels of the organization.

Training Objectives

- ✔ Ensure personnel understand their roles and responsibilities during BCP activation
- ✔ Promote awareness of critical business functions and associated recovery protocols

Training Modules & Delivery

Training Type	Target Audience	Frequency	Delivery Mode
BCP Induction Training	All new employees	Upon joining	LMS module + live orientation
Annual Refresher Training	Critical departments (QA, Production, IT, Engineering, Regulatory)	Annually	Department-led sessions + LMS
Scenario-Specific Training	Incident Response Team (IRT), BCP Leads	Quarterly / As needed	Tabletop exercises, drills
Post-Test Debriefing Sessions	All test participants	After each BCP test	Facilitated workshop + CAPA review

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Training Type	Target Audience	Frequency	Delivery Mode
Emergency Response SOP Training	EHS, Security, Engineering	Semi-annually	Hands-on drills + SOP walkthrough
Crisis Communication Protocols	Site Heads, Spokespersons, HR	Annually	Role-play simulations + SOP review

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ANNEXURES:

Annexure 1 - Critical Business Functions Matrix.

Annexure 2 - Risk Assessment & Impact Analysis Matrix.

Annexure 3 - Emergency Contact List.

Annexure 4 - List of Critical Equipment.

Annexure 5 - Alternate Supplier and Service Provider List (Refer current Approved List)

Annexure 6 - Emergency response plan (Refer current Approved SOP)

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Revision History – Business Continuity Plan (BCP)

Document No.	Version No.	Effective Date	Nature of Revision	Reason for Change

Note: All revisions are reviewed and approved by the BCP Committee and documented in the QMS Change Control Log. Superseded versions are archived per document control SOP.

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Annexure 1 - Critical Business Functions Matrix

Department	Critical Business Function (CBF)	Description	Criticality Level	Recovery Time Objective (RTO)	Responsible Team(s)
Production – Upstream	Fermentation & Media Preparation	Cell culture, media prep, inoculation, fermentation	High	≤ 4 hours	Production, QA, Engineering
Production – Downstream	Chromatography & Filtration	Purification, buffer prep, UF/DF, diafiltration	High	≤ 6 hours	Production, QA, QC, Engineering
Production – Fill-Finish	Aseptic Filling & Inspection	Sterile filling, visual inspection, labelling	High	≤ 2 hours	Production, QA, Regulatory Affairs
Quality Assurance (QA)	Batch Release & QMS Oversight	GMP compliance, deviation handling, document control	High	≤ 8 hours	QA, Regulatory Affairs
Quality Control (QC)	Analytical & Stability Testing	Raw material, in-process, finished product testing	High	≤ 12 hours	QC, QA, Regulatory Affairs
Microbiology	Environmental Monitoring & Sterility	EM trending, sterility testing, microbial ID	High	≤ 6 hours	QC Microbiology, QA

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Department	Critical Business Function (CBF)	Description	Criticality Level	Recovery Time Objective (RTO)	Responsible Team(s)
Regulatory Affairs	Dossier Preparation & Submission	Regulatory filings, query responses, compliance alignment	High	≤ 24 hours	Regulatory Affairs, QA, RA SMEs
R&D	Tech Transfer & Process Development	Scale-up, validation, documentation, regulatory support	High	≤ 48 hours	R&D, QA, RA, Manufacturing
Engineering & Utilities	HVAC, Water, Steam, Power Systems	Utility continuity, equipment readiness, preventive maintenance	High	≤ 1 hours	Engineering, QA, EHS
IT & Data Integrity	GxP Systems & Cybersecurity	DMS, LMS, SCADA, backups, access control	High	≤ 1 hour	IT, QA, Data Integrity
Warehouse & Logistics	Inventory & Cold Chain Management	Material movement, temperature control, dispatch	High	≤ 4 hours	Warehouse, QA, Supply Chain
HR & Administration	Workforce Continuity & Safety	Staffing, training, emergency response, POSH	Medium	≤ 24 hours	HR, Admin, QA
Safety & EHS	Emergency Response & Fire/Flood Control	Fire drills, spill control, evacuation, containment	High	Immediate	EHS, Engineering, Security

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Department	Critical Business Function (CBF)	Description	Criticality Level	Recovery Time Objective (RTO)	Responsible Team(s)
Corporate Affairs	Crisis Communication & Stakeholder Mgmt	Internal/external messaging, media handling, regulatory liaison	Medium	≤ 6 hours	Corporate Affairs, QA, Regulatory Affairs

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Annexure 2 : Risk Assessment & Impact Analysis Matrix

Risk ID	Risk Scenario	Impacted CBFs	Severity	Likelihood	Notes / Mitigation Strategy
R-001	Power outage due to grid failure or internal fault	HVAC, Water, Steam, Power Systems; Fermentation, Fill-Finish, IT Systems	High	Medium	Backup generators, UPS for IT, preventive maintenance, utility, RTO ≤ 2 hrs
R-002	Contamination during aseptic filling	Aseptic Filling & Inspection; Batch Release	High	Low	Enhanced EM, periodic sterility audits, operator retraining, RTO ≤ 2 hrs
R-003	Cyberattack on GxP systems	GxP Systems & Cybersecurity; QMS Oversight; Dossier Submission	High	Medium	Multi-layered access control, SCADA isolation, daily backups, RTO ≤ 1 hr
R-004	HVAC failure during fermentation	Fermentation & Media Prep, EM & Sterility	High	Medium	Redundant HVAC units, alarm systems, Engineering SOPs, RTO ≤ 4 hrs
R-005	Delay in regulatory query response	Dossier Preparation & Submission; Batch Release	Medium	Medium	RA SME pool, pre-approved response templates, RTO ≤ 24 hrs
R-006	Fire in warehouse or production block	Inventory & Cold Chain, Chromatography, Fill-Finish	High	Low	Fire detection, suppression systems, evacuation drills, RTO ≤ 4 hrs
R-007	Cold chain breach during dispatch	Inventory & Cold Chain, Stability Testing	High	Medium	Real-time temperature monitoring, validated packaging, RTO ≤ 4 hrs

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Risk ID	Risk Scenario	Impacted CBFs	Severity	Likelihood	Notes / Mitigation Strategy		
R-008	Staff unavailability due to pandemic or strike	Workforce Continuity; All Production CBFs	High	Medium	Cross-training, remote access, HR emergency protocols, RTO ≤ 24 hrs		
R-009	Equipment failure in chromatography	Chromatography & Filtration; Analytical Testing	High	Medium	Spare parts inventory, vendor SLA, Engineering response SOP, RTO ≤ 6 hrs		
R-010	Data integrity breach in QMS	QMS Oversight, Batch Release, Regulatory Submission	High	Low	Audit trails, restricted access, periodic DI audits, RTO ≤ 8 hrs		
R-011	Flooding due to monsoon or plumbing failure	Emergency Response; Utilities; Warehouse	High	Medium	Drainage audits, containment kits, EHS SOPs, RTO = Immediate		
R-012	Negative media coverage or stakeholder panic	Crisis Communication; Regulatory Affairs	Medium	Medium	Pre-drafted media statements, stakeholder briefings, RTO ≤ 6 hrs		
R-013	Delay in tech transfer validation	Tech Transfer & Process Development; Manufacturing	Medium	Medium	Parallel documentation, validation templates, RTO ≤ 48 hrs		

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Annexure 3 – Emergency Contact Numbers**A. Site Emergency Contacts**

Sr. No.	Agency / Individual	Contact Number(s)	Remarks
1	Government Ambulance		24x7 Emergency Medical Response
2	Ampada Veedu Ambulance		Local Ambulance Service
3	Mahatma Gandhi Medical College & Research Inst.		Tertiary Care Hospital
4	Fire Service		National Emergency Fire Response
5	Fire Station – Bahour		Local Fire Station
6	Electricity Board (EB)		Local Grid Support
7	Emergency Support System		Integrated National Helpline
8	District Collector		Administrative Head
9	Superintendent of Police		Law Enforcement
10	District Revenue Officer		Revenue & Land Administration
11	Thasildhar		Local Revenue Officer

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Sr. No.	Agency / Individual	Contact Number(s)	Remarks
12	Inspector		Police Emergency
13	National Cyber Crime Helpline		Cybersecurity Incident Reporting
14	Chief Inspector of Factories		Industrial Safety Oversight
15	Pollution Control Board		Environmental Compliance
16	Pollution Control Board		Regional Environmental Oversight
17	Inspector of Boilers		Boiler Safety Compliance
18	Labour Commissioner		Labour Law & Welfare
19-30	Site HODs & Internal Contacts	See internal directory below	Departmental Emergency Coordination

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Internal Contacts – Site

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B. R&D Facility Emergency Contacts

Sl. No.	Agency / Individual	Contact Number(s)	Remarks
1	Government Ambulance		24x7 Emergency Medical Response
2	Fire Service		National Emergency Fire Response
3	Fire Station		Local Fire Station
4	Electricity Board		Grid Emergency Support
5	Emergency Support System		Integrated National Helpline
6	District Collector		Administrative Head
7	Superintendent of Police		Law Enforcement
8	Chief Inspector of Factories		Industrial Safety Oversight
9	Maharashtra Pollution Control Board		Environmental Compliance
10-24	Internal Contacts	<i>To be updated</i>	Recommend completion for audit

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Corporate Office Emergency Contacts

Sr. No.	Agency / Individual	Contact Number(s)	Remarks
1	Government Ambulance		24x7 Emergency Medical Response
2	Fire Service		National Emergency Fire Response
3	Fire Station		Local Fire Station
4	Electricity Board		Grid Emergency Support
5	Emergency Support System		Integrated National Helpline
6	District Collector		Administrative Head
7	Superintendent of Police		Law Enforcement
8	Chief Inspector of Factories		Industrial Safety Oversight
9	Maharashtra Pollution Control Board		Environmental Compliance
10	Labour Commissioner		Labour Law & Welfare

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Annexure 4 – Departmental Critical Equipment List**A: Formulation Department – Critical Equipment**

Sr. No.	Equipment Name	Equipment ID	Location	Function / Criticality

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B: DS Department – Critical Equipment

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C: Purification Department – Critical Equipment

[illegible]

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D: Engineering Department – Utility Systems

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E: Warehouse – Storage & Sampling Equipment

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F: Quality Control – Analytical Instruments

Sr. No.	Instrument Name	Instrument ID	Location	Function / Criticality

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C: Microbiology

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